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VTU Previous Question Papers

BE - CS

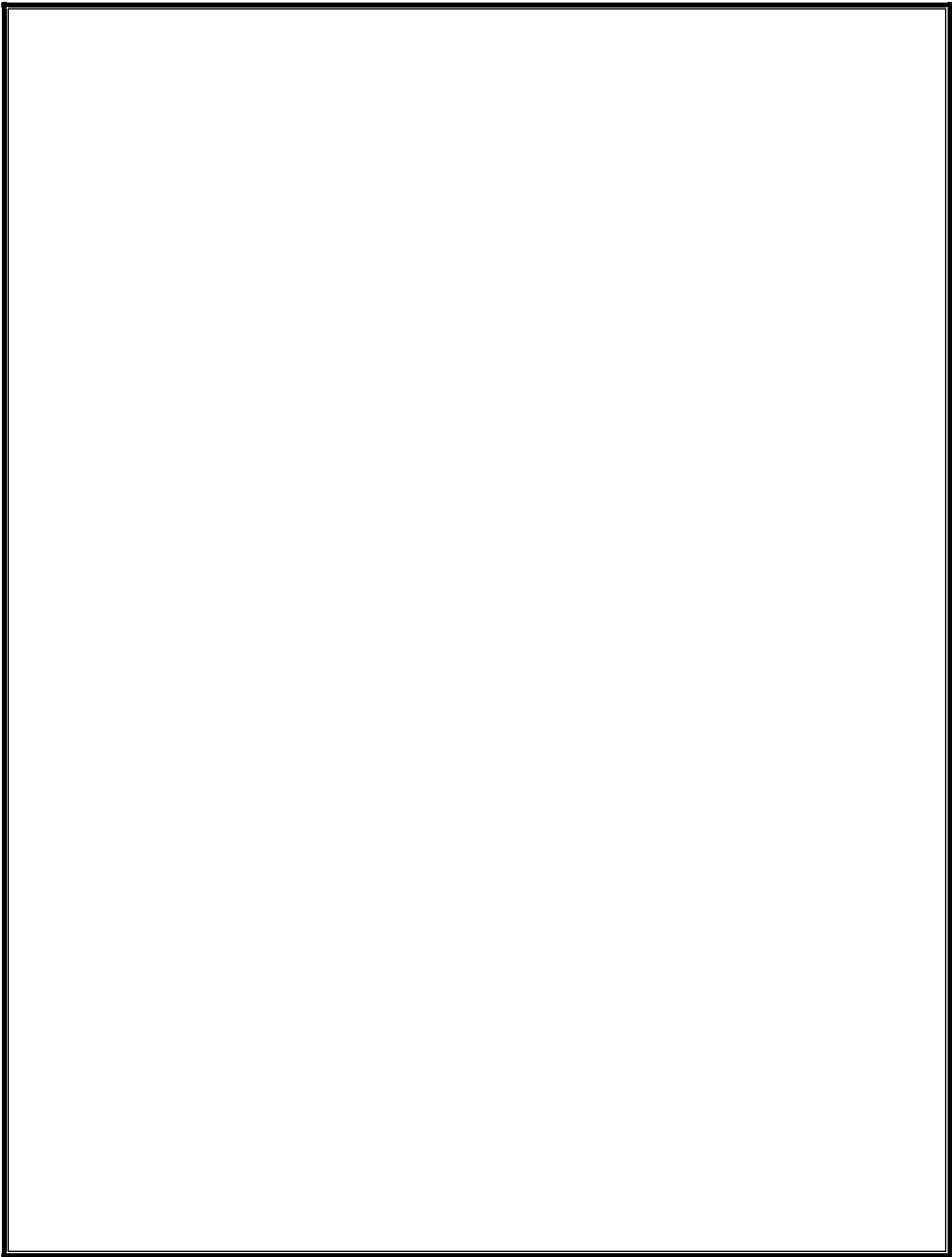
III to VIII Semester

Jun/Jul-2024

2018 & 2022 Scheme

VVIET

Bannur road, Alanahalli, Mysore-570028



Sl.No.	Sub-Code	Subject Title	Exam Date
1	18CPS13/23	C -programming for Problem Solving	June/July 2024
2	18CS32	Data Structures and Applications	June/July 2024
3	18CS33	Analog and Digital Electronics	June/July 2024
4	18CS34	Computer Organization	June/July 2024
5	18CS35	Software Engineering	June/July 2024
6	18CS36	Discrete Mathematical Structures	June/July 2024
7	18CS42	Design and Analysis of Algorithms	June/July 2024
8	18CS46	Data Communication	June/July 2024
9	18CS51	Management and Entrepreneurship for IT Industry	June/July 2024
10	18CS52	Computer Networks and Security	June/July 2024
11	18CS53	Database Management Systems	June/July 2024
12	18CS54	Automata Theory and Computability	June/July 2024
13	18CS55	Application Development using Python	June/July 2024
14	18CS81	Internet of Things	June/July 2024
15	18CS641	Data Mining and Data Warehousing	June/July 2024
16	18CS642	Object Oriented Modelling and Design	June/July 2024
17	18CS734	User Interface Design	June/July 2024
18	18CS741	Digital Image Processing	June/July 2024
19	18CS823	NOSQL Database	June/July 2024
20	21CS32	Data Structures and Applications	June/July 2024
21	21CS33	Analog and Digital Electronics	June/July 2024

22	21CS42	Design and Analysis of Algorithms	June/July 2024
23	21CS44	Operating Systems	June/July 2024
24	21CS51	Automata Theory and Compiler Design	June/July 2024
25	21CS53	Database Management System s	June/July 2024
26	21CS61	Software Engineering and Project Management	June/July 2024
27	21CS62	Full Stack Development	June/July 2024
28	21CS63	Computer Graphics and Fundamental s of Image Processing	June/July 2024
29	21CS482	Unix Shell Programming	June/July 2024
30	21CS642	Advanced Java Programming	June/July 2024
31	BBOC407	Biology for Engineers (CSE)	June/July 2024
32	BBOK407	Biology for Engineers	June/July 2024
33	BCS301	Mathematics for Computer Science	June/July 2024
34	BCS302	Digital Design and Computer Organization	June/July 2024
35	BCS303	Operating Systems	June/July 2024
36	BCS304	Data Structures and Applications	June/July 2024
37	BCS306A	Object Oriented Programming with Java	June/July 2024
38	BCS401	Analysis and Design of Algorithms	June/July 2024
39	BCS402	Microcontroller s	June/July 2024
40	BCS403	Database Management Systems	June/July 2024
41	BCS405A	Discrete Mathematical Structures	June/July 2024
42	BCS405B	Graph Theory	June/July 2024

CBCS SCHEME

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18CPS13/23

First/Second Semester B.E. Degree Examination, June/July 2024 C-programming for Problem Solving

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the following :
 - i) Generations of computer
 - ii) Computer in a network. (08 Marks)
- b. Define data type? Explain various data types available in C with an example. (08 Marks)
- c. Identify whether the given variable names are valid or invalid with a suitable reason.
 - i) First_tag ii) Char iii) group one iv) int_type. (04 Marks)

OR

- 2 a. Explain the basic structure of C program with an example. (08 Marks)
- b. Write a note on operator precedence and associativity. (08 Marks)
- c. Evaluate the following expressions :
 - i) $100\% 20 <= 20 - 5 + 100\% 10 - 20 == 5 >= 1! = 20$
 - ii) $a + = b * = c - = 5$ where $a = 3$, $b = 5$ and $c = 8$. (04 Marks)

Module-2

- 3 a. Explain the following with syntax and an example : i) if else ii) switch. (08 Marks)
- b. Write a C program to compute the roots of a quadratic equation by accepting the coefficients and print appropriate messages. (08 Marks)
- c. Explain the following : i) goto ii) break. (04 Marks)

OR

- 4 a. Write a C program for plotting Pascal of a triangle. (08 Marks)
- b. Explain the formatted output statement with an example. (08 Marks)
- c. Differentiate between while and do-while loop. (04 Marks)

Module-3

- 5 a. Explain the declaration and initialization of one dimensional array with an example. (08 Marks)
- b. List all the string manipulation functions and explain string comparison and string copy functions with an example. (08 Marks)
- c. Write a note on how a list of names can be stored in a two dimensional character array. (04 Marks)

OR

- 6 a. Write a program to sort the given set of N numbers using bubble sort. (08 Marks)
- b. Explain the declaration and different way of initializing two dimensional array with an example for each. (08 Marks)
- c. Write a program to copy one string into another and count the number of characters copied. (04 Marks)

Module-4

- 7 a. Explain in detail pass by value and pass by reference with an example for each. (08 Marks)
b. Explain the different categories of function with an example for each. (08 Marks)
c. Write a C program to convert binary number to decimal number using recursion. (04 Marks)

OR

- 8 a. Explain the following :
i) Function definition
ii) Function declaration
iii) Function call. (08 Marks)
b. What is recursion? Write a program to find the factorial of a given number using recursion concept. (08 Marks)
c. Write a program to swap two numbers by using call by reference. (04 Marks)

Module-5

- 9 a. Define structure? Explain the declaration and initialization of structure with an example. (08 Marks)
b. Explain the concept of pointer to pointer with an example. (08 Marks)
c. Write a note on array of structures. (04 Marks)

OR

- 10 a. Write a program to read, write and compute the average marks and the students solving above and below the average marks for a class of N students using structures. (08 Marks)
b. What is a pointer? Explain declaration and initialization of pointer with an example. (08 Marks)
c. Explain the following preprocessor directives. i) #define ii) #error. (04 Marks)

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18CS32

Third Semester B.E. Degree Examination, June/July 2024 Data Structures and Applications

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Data Structures. Give its classification. What are the basic operations that can be performed on Data structure? (07 Marks)
- b. Give the ADT for sparse matrix. Express the given sparse matrix in the triplet form and find its transpose.

$$A = \begin{bmatrix} 30 & 0 & 0 & 25 & 0 \\ 0 & 0 & 40 & 0 & 22 \\ 0 & 0 & 0 & 0 & 42 \\ 52 & 0 & 0 & 61 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 10 & 0 & 0 \end{bmatrix}$$

(07 Marks)

- c. Write a program to search for key element in an array using binary search technique.

(06 Marks)

OR

- 2 a. Define strings. List and explain any 4 operations with example. (10 Marks)
- b. Write a program to implement a C program for the following array operations:
 - (i) Creating an array of N integer elements.
 - (ii) Display of array elements with suitable headings.
 - (iii) Inserting an element at a given valid positions
 - (iv) Deleting an element at a given valid positions.
 Support the program with functions for each of the above operations. (10 Marks)

Module-2

- 3 a. Define a stack. Write an algorithm to perform different operations on stack and then using diagrammatic representation. (10 Marks)
- b. Convert the following infix expression to postfix expression,
 - (i) $((A + (B - C) * D) \wedge E + F)$
 - (ii) $X\$Y\$Z - M + N + \frac{P}{Q}$(06 Marks)
- c. Write a program to find GCD of 2 integer numbers. Using recursion. (04 Marks)

OR

- 4 a. Define a queue. Write a program to implement Qinsert (), Qdisplay (), Qdelete () in C using arrays. (10 Marks)
- b. Write an algorithm for evaluating a valid postfix expression. Trace the same on $562 + *841 - + *$. (07 Marks)
- c. Differentiate recursion and iteration process. (03 Marks)

Module-3

- 5 a. With the C-statements, explain how do you create a node, add and delete on singly linked list (SLL) with proper messages where each node containing the details of an employee in the form of Empid, empage, empname, empsalary as data fields. (10 Marks)
- b. Write C functions for the following operations on Doubly linked list (DLL) :
 (i) Concatenation of two DLL
 (ii) Search the DLL for the given key element. (10 Marks)

OR

- 6 a. Explain the following with suitable example: (i) Circular linked list (ii) Doubly linked list (10 Marks)
- b. Write an algorithm to add 2 polynomials using circular singly linked list and also represent one example of polynomial using CSLL. (10 Marks)

Module-4

- 7 a. Define the following tree terminologies with example :
 (i) Degree of a node
 (ii) Strictly Binary tree
 (iii) Level of a binary tree
 (iv) Siblings
 (v) Skewed BT (10 Marks)
- b. Write recursive functions for in-order, pre-order, post-order traversal of binary tree. Also give the 3 traversal for the BT given below Fig.Q7 (b). (10 Marks)

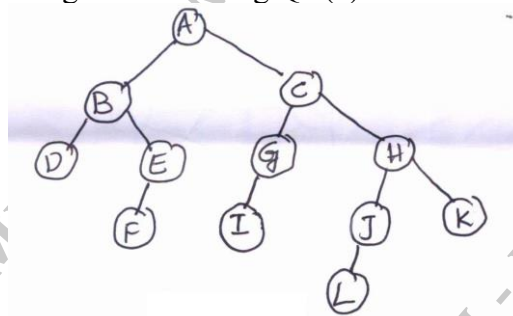


Fig. Q7 (b)

OR

- 8 a. Define Binary Search Tree (BST). Construct BST for the element step-by-step. 100, 85, 45, 55, 110, 20, 70, 65, 113, 145, 132, 96 (10 Marks)
- b. Construct an expression tree for the given expression and traverse in preorder and postorder for the same.
 $((6 + (3 - 2) * 5) \wedge 2 + 3)$ (10 Marks)

Module-5

- 9 a. Define the following terms : (i) Graph (ii) Multigraph (iii) Loop (iv) Subgraph (04 Marks)
- b. What is Hashing? Explain the following hashing functions :
 (i) Division method
 (ii) MidSquare method
 (iii) Folding method. (08 Marks)
- c. Write a C program to sort the elements in increasing order using insertion sort technique. (08 Marks)

OR

- 10 a. What is file? List basic file operations. Explain any four operations with syntax and example. (08 Marks)
- b. Write the adjacency matrix and adjacency list representation for the given graph in Fig. Q10 (b). (06 Marks)

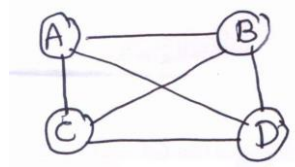


Fig. Q10 (b)

- c. Explain with example the Radix sort technique and trace it. Arrange the numbers in ascending order with 2 and 3 digit numbers. (06 Marks)

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Third Semester B.E. Degree Examination, June/July 2024 Analog and Digital Electronics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the construction working and characteristics of Light Emitting Diode. (06 Marks)
- b. With a neat circuit diagram and Mathematical analysis explain fixed bias circuit. (07 Marks)
- c. Show how IC-555 timer can be used as Astable Multivibrator with Waveforms. (07 Marks)

OR

- 2 a. Discuss successive approximation register method of A to D converter with detailed conversion process. (08 Marks)
- b. With neat diagram and waveform, explain working of inverting Schmitt trigger circuit. (06 Marks)
- c. Explain Adjustable Voltage Regulator with diagram and suitable equations. (06 Marks)

Module-2

- 3 a. Simplify the following function using K-map and obtain simplified Boolean expressions:
 $f_1(a, b, c, d) = \sum m(0, 1, 2, 4, 5, 6, 8, 9, 10, 12, 13)$
 $f_2(a, b, c, d) = \sum m(1, 3, 5, 7, 9) + \sum d(6, 12, 13)$. (10 Marks)
- b. Simplify the following Boolean function by using Quine-Mcclusky (QM) method
 $F(A, B, C, D) = \sum m(0, 2, 3, 6, 7, 8, 10, 12, 13)$. Find all the prime implicants. (10 Marks)

OR

- 4 a. Minimize the following function using MEV technique, use 'd' a MEV variable
 $f(A, B, C, D) = \sum m(0, 1, 2, 7, 8, 9, 14, 15)$. (08 Marks)
- b. With an example, explain Petrik's method. (06 Marks)
- c. Solve the following clearly mention prime implicants and essential prime implicants
 $f(a, b, c, d) = \sum m(1, 5, 6, 7, 11, 12, 13, 15)$. (06 Marks)

Module-3

- 5 a. What are Hazards in digital systems? Explain static 1 and static – 0 hazards. (08 Marks)
- b. What is Multiplexer? Discuss 8 to 1 MUX with the help of logic diagram and equation. (06 Marks)
- c. Discuss the importance of three state buffer with example. (06 Marks)

OR

- 6 a. Show how using a 3 to 8 decoder and multiinput or gates following Boolean expressions can be realized simultaneously
 $F_1(A, B, C) = \sum m(0, 4, 6)$
 $F_2(A, B, C) = \sum m(0, 5)$
 $F_3(A, B, C) = \sum m(1, 2, 3, 7)$ (06 Marks)
- b. Realize $f(a, b, c, d) = \sum m(1, 5, 6, 10, 13, 14)$ using AND-OR logic with number of levels, Gates and Gate inputs. (06 Marks)
- c. Write a short note on PLA and PAL. 3 of 3 (08 Marks)

Module-4

- 7 a. Discuss the operation of SR-Latch using NOR gates. Show how SR Latch can be used for switch debouncing. (08 Marks)
b. Explain Gated SR-latch using NAND gate. (06 Marks)
c. Differentiate between Latch and flip flop and explain the structure of VHDL program. (06 Marks)

OR

- 8 a. Explain the working operation of SR-flipflop and JK flip flop with truth table and waveforms. (08 Marks)
b. Draw the logic diagram of master slave JK flip flop using NAND gates and explain the working with suitable timing diagram. (07 Marks)
c. Discuss Toggle Flip Flop with truth table and characteristic equation. (05 Marks)

Module-5

- 9 a. Explain Parallel Adder with Accumulator with neat diagram and operation. (08 Marks)
b. What is Register? Explain how 4 bit register with data load clear and clock constructed using D flip flops. (07 Marks)
c. Discuss the operation of data transfer between Register. (05 Marks)

OR

- 10 a. Design the counter using D flip flop for the given sequence 0 – 3 – 2 – 6 – 4 – 7 – 0. (08 Marks)
b. Explain synchronous Binary Counter with logical diagram and transition table. (06 Marks)
c. Explain the working of 8 bit serial - in – serial - out shift register using SR flip flop. (06 Marks)

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Third Semester B.E. Degree Examination, June/July 2024 Computer Organization

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a neat diagram, analyze the basic operational concept of a computer. Give the operational steps (operational steps). (10 Marks)
- b. What is performance measurement? Explain the overall SPEC rating for the computer in a program suite. (10 Marks)

OR

- 2 a. Define addressing mode, explain the various addressing mode with examples. (10 Marks)
- b. Explain in detail, various shift and rotate instruction and example with neat diagram. (10 Marks)

Module-2

- 3 a. With neat diagram, explain various methods for handling multiple interrupts requests raised by multiple devices. (10 Marks)
- b. What is DMA Bus Arbitration? Briefly explain different bus arbitration techniques. (10 Marks)

OR

- 4 a. What is an interrupt? What are interrupt service routine and what are vector interrupts? Explain with example. (10 Marks)
- b. Explain the following with respect to USB: (10 Marks)
 - i) U.S.B. Architecture
 - ii) U.S.B. protocols.

Module-3

- 5 a. Draw a diagram and explain the working of 16 megabit DRAM chip configured as $2m \times 8$. (10 Marks)
- b. Explain direct mapping technique and associative mapping technique in mapping function. (10 Marks)

OR

- 6 a. Analyze how data are written into (ROM) Read Only Memory. Discuss different types of Read Only Memories. (10 Marks)
- b. What is cache memory? Explain any two mapping functions of cache memory. (10 Marks)

Module-4

- 7 a. Convert the following pairs of decimal numbers to 5-bit signed 2's complement binary numbers and add them. State whether or not over flow occurs in each case: (04 Marks)
 - i) 5 and 10
 - ii) -14 and 11
 - iii) -5 and 7
 - iv) -10 and -13

- b. Design the 16 bit carry look ahead adder using 4-bit adder. Also write the expression for C_{i+1} . (10 Marks)
- c. Draw the two n-bit number x and y to perform addition/subtraction. (06 Marks)

OR

- 8 a. With an example explain the Booths algorithm to multiply two signed operands. (10 Marks)
- b. Multiply each of the following pairs of signed 2's complement number using the booth algorithm (A = multiplicand and B = multiplier)
- A = 010111 and B = 110110
 - A = 110011 and B = 101100
 - A = 110101 and B = 011011
 - A = 001111 and B = 001111
- (10 Marks)

Module-5

- 9 a. Discuss with neat diagram, the single bus organization of the data path inside a processor. (10 Marks)
- b. Write the sequence of control steps required for single bus structure for each if the following instructions.
- Add the contents of memory location NUM to register R1.
 - Add the contents of memory location whose address is at memory location NUM to register R1.
- (10 Marks)

OR

- 10 a. Explain the following:
- Hard-wired control
 - Microprogrammed control.
- (10 Marks)
- b. What is pipeline? Explain the 4 stages pipeline with its instruction execution steps and hardware organization. (10 Marks)

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Third Semester B.E. Degree Examination, June/July 2024 Software Engineering

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Software Engineering? Explain the waterfall model approach to develop a software product with a neat diagram. (10 Marks)
- b. Explain Insulin pump control system with a neat diagram. (10 Marks)

OR

- 2 a. Explain Requirement Engineering process with a neat diagram. (10 Marks)
- b. In connection with requirements analysis and elicitation, explain the following:
i) Scenario ii) Interviewing (10 Marks)

Module-2

- 3 a. What is object orientation? Explain the four aspects or characteristics required by an object oriented approach. (10 Marks)
- b. What is an object oriented development? Explain object oriented themes. (10 Marks)

OR

- 4 a. Explain the following:
i) Link
ii) Association
iii) Multiplicity
iv) Ordering (10 Marks)
- b. Explain association classes and qualified associations with an example for each. (10 Marks)

Module-3

- 5 a. What are structural models? In connection with structural models, explain the following:
i) Generalization ii) Aggregation (10 Marks)
- b. What is Event-Driven modeling? Explain the state diagram of a microwave oven. (10 Marks)

OR

- 6 a. Explain the Rational Unified process, with a neat diagram. (10 Marks)
- b. What are Design Models? Explain weather station state diagram. (10 Marks)

Module-4

- 7 a. What is Software Testing? Explain Test Driven Development with a neat diagram. (10 Marks)
- b. What is Acceptance Testing? Explain Acceptance Testing process with a neat diagram. (10 Marks)

OR

- 8 a. What is Software Evolution? Explain software evolution process with a neat diagram. (10 Marks)
b. What is Software Re-Engineering? Explain software evolution process with a neat diagram. (10 Marks)

Module-5

- 9 a. What is Project Planning? Explain the project planning process with a neat diagram. (10 Marks)
b. Explain the following:
i) Project scheduling
ii) Estimation techniques (10 Marks)

OR

- 10 a. What is Software Quality? With a neat diagram, explain process based approach to achieve product quality. (10 Marks)
b. What is Software Review? Explain the software Review process with a neat diagram. (10 Marks)

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18CS36

Third Semester B.E. Degree Examination, June/July 2024 Discrete Mathematical Structures

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following :
 - (i) Proposition
 - (ii) Tautology
 - (iii) Contradiction

Verify whether the following compound proposition is a Tautology

$$[(p \rightarrow q) \wedge (q \rightarrow r)] \rightarrow (p \rightarrow r) \quad (07 \text{ Marks})$$
- b. By constructing the Truth Table show that,

$$[(p \vee q) \rightarrow r] \Leftrightarrow [(p \rightarrow r) \wedge (q \rightarrow r)] \quad (07 \text{ Marks})$$
- c. Let $p(x) : x^2 - 7x + 10 = 0$, $q(x) : x^2 - 2x + 3 = 0$, $r(x) : x < 0$
Find the truth or falsity of the following statements, when the universe U contains only the integers 2 and 5.
 - (i) $\forall x, p(x) \rightarrow \neg r(x)$
 - (ii) $\forall x, q(x) \rightarrow r(x)$
 - (iii) $\exists x q(x) \rightarrow r(x)$
 - (iv) $\exists x p(x) \rightarrow r(x)$ (06 Marks)

OR

- 2 a. Define Converse, Inverse and Contra-positive of a conditional. Represent these in the form of a truth table. (07 Marks)
- b. For each of the following statements, provide an indirect proof by stating and proving the contrapositive of the given statement.
 - (i) For all integers k and ℓ , if k ℓ is odd, then both k and ℓ are odd.
 - (ii) For all integers k and ℓ , if $k + \ell$ is even, then k and ℓ are both even and odd. (07 Marks)
- c. (i) I will get grade A in this course or I will not graduate.
If I do not graduate, I will join the army
I got grade A.

$\therefore$ I will not join the army.
Is this a valid argument?
- (ii) If Ravi studies, he will pass in Discrete Mathematics paper
If Ravi does not play PUBG, then he will study,
Ravi failed in Discrete Mathematics paper

$\therefore$ Ravi played PUBG
Check the validity of this argument. (06 Marks)

Module-2

- 3 a. Prove by mathematical induction that, for every positive integer n , 5 divides $n^5 - n$. (06 Marks)
- b. For all positive integers n , prove that, if $n \geq 24$, then n can be written as a sum of 5's and 7's. (07 Marks)
- c. Define the well ordering principle, by using mathematical induction principle, prove that
- $$\frac{1}{2.5} + \frac{1}{5.8} + \frac{1}{0.11} + \dots + \frac{1}{(3n-1)(3n+2)} = \frac{n}{6n+4}.$$
- (07 Marks)

OR

- 4 a. Prove by mathematical induction that, for every positive integer n , the number $11^{n+2} + 12^{2n+1}$ is divisible by 133. (06 Marks)
- b. Determine the coefficient of,
- (i) $x^{11}y^4$ in the expansion of $(2x^3 - 3xy^2 + z^2)^6$.
- (ii) $a^2b^3c^2d^5$ in the expansion of, $(a + 2b - 3c + 2d + 5)^{16}$ (07 Marks)
- c. In how many ways can one distribute eight identical balls into four distinct containers so that,
- (i) No container is empty?
- (ii) The fourth container gets an odd number of balls? (07 Marks)

Module-3

- 5 a. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined by $f(x) = \begin{cases} 3x-5 & \text{for } x > 0 \\ 3x+5 & \text{for } x \leq 0 \end{cases}$.
- (i) Determine $f(0)$, $f(-1)$, $f\left(\frac{3}{5}\right)$, $f\left(-\frac{3}{5}\right)$.
- (ii) Find $f^{-1}[(-5,5)]$ (07 Marks)
- b. Let $A = \{1, 2, 3, 4\}$, $B = \{a, b, c\}$, $C = \{3, 4, 7\}$. Write down the following $A \cup (B \times C)$, $(A \cup B) \times C$, $(A \times C) \cup B \times C$. (07 Marks)
- c. Let f, g, h be functions from $\mathbb{Z}$ to $\mathbb{Z}$ defined by $f(x) = x - 1$, $g(x) = 3x$,
 $h(x) = 0$ if x is even
 $= 1$ if x is odd.
- Determine $(f \circ (g \circ h))(x) = ((f \circ g) \circ h)(x)$ and verify $(f \circ (g \circ h)) = ((f \circ g) \circ h)$. (06 Marks)

OR

- 6 a. State pigeonhole principle. Let $\triangle ABC$ be an equilateral triangle of side 1 cm each. Show that if we select 10 points in the interior, there must be at least two points whose distance is less than $\frac{1}{2}$ cm. (07 Marks)
- b. Draw Hasse diagram representing the positive divisors of 36. (07 Marks)
- c. Let $A = \{1, 2, 3\}$ and f, g, h, p be functions on A defined as follows :
- $$f = \{(1,2)(2,3)(3,1)\}, g = \{(1,2)(2,1)(3,3)\}, h = \{(1,1)(2,2)(3,1)\}, p = \{(1,1)(2,2)(3,2)\}$$
- Find $f \circ g, g \circ f, f \circ p, p \circ g, g \circ p, f \circ h \circ g$ (06 Marks)

Module-4

- 7 a. Define derangement. (07 Marks)
- There are eight letters to eight different people to be placed in eight different addressed envelopes. Find the number of ways of doing this so that at least one letter goes to the right person.
 - Find the number of derangement of 1, 2, 3, 4.
- b. Find the rook polynomial for the 3×3 board by using the expansion formula. (07 Marks)
- c. Define homogeneous and non-homogeneous recurrence relations of first order and solve the recurrence relation.
 $a_n = 5a_{n-1} + 6a_{n-2}$, $n \geq 2$, $a_0 = 1$, $a_1 = 5$. (06 Marks)

OR

- 8 a. Solve the following :
- From seven consonants and five vowels how many sets consisting of four different consonants and three different vowels can be formed?
 - Find the number of arrangement of the letters in TALLAHASSE. Which have no adjacent A's. (07 Marks)
- b. How many integers between 1 and 300 (inclusive) are,
- Divisible by 5, 6, 8?
 - Divisible by none of 5, 6, 8? (07 Marks)
- c. An apple, a banana, a mango and an orange are to be distributed to four boys B_1 , B_2 , B_3 and B_4 . The boys B_1 and B_2 do not wish to have apple, the boy B_3 does not want banana or mango and B_4 refuses orange. In how many ways the distribution can be made so that no boy is displeased? (06 Marks)

Module-5

- 9 a. Define the following terms with respect to graph :
- Directed graph
 - A walk
 - Sub-graph
 - Connected graph
 - Simple graph
 - Regular graph
 - Complete graph (07 Marks)
- b. Define complete bipartite graph. How many vertices and how many edges are there in $K_{4,7}$ and $K_{7,11}$? (07 Marks)
- c. Let $G = (V, E)$ be the undirected graph in the Fig. Q9 (c). How many paths are there in G from a to h? How many of these paths have a length 5? (06 Marks)

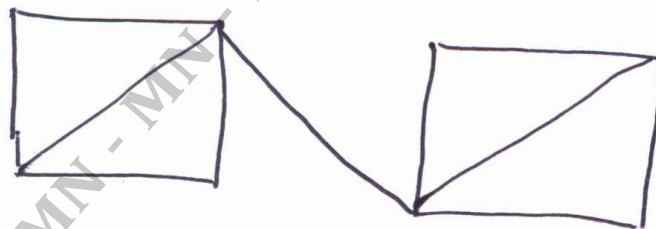


Fig. Q9 (c)

OR

- 10 a. Define a Tree. Prove that the tree $G = (V, E)$ with P vertices has $(P - 1)$ edges. (07 Marks)
- b. Define a spanning tree of a graph. Find all the spanning trees of the following graph in Fig.Q10 (b). (07 Marks)

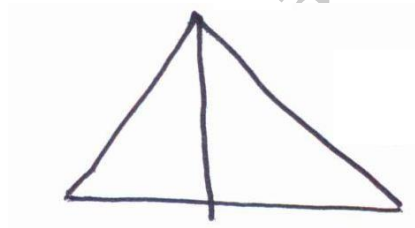


Fig. Q10 (b)

- c. Obtain an optimal prefix code for the message LETTER RECEIVED using labelled binary tree. Indicate the code. (06 Marks)

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Fourth Semester B.E. Degree Examination, June/July 2024
Design and Analysis of Algorithms

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define an algorithm. Design an algorithm to search an element in an array of elements using sequential search. Discuss worst, best and average efficiency of the algorithm. (08 Marks)
- b. Explain the asymptotic notations: i) Big-oh (O) ii) Big omega (Ω) iii) Theta (θ) with an example for each. Also prove that: If $t_1(n) \in O(g_1(n))$ and $t_2(n) \in O(g_2(n))$ then $t_1(n) + t_2(n) \in O(\max \{g_1(n), g_2(n)\})$. (08 Marks)
- c. Discuss adjacency matrix and adjacency list representation of a graph with suitable examples. (04 Marks)

OR

- 2 a. Write a recursive algorithm to print all the permutations of a set of $n \geq 1$ elements. Also write a recursive tree of calls for $n = 3$. (08 Marks)
- b. Give a general plan for analyzing recursive algorithms. Give a recursive algorithm to find the number of binary digits in the binary representation of a positive decimal integer and obtain its efficiency. (08 Marks)
- c. Explain with an example, how to convert ordered rooted tree into a binary tree. (04 Marks)

Module-2

- 3 a. Given a bag of 16 coins and let seventh one be the counterfeit coin among 16 coins. Counterfeit coins are lighter than genuine ones. Given a machine to compare the weights of two sets of coins, apply divide and conquer to determine that 7th is counterfeit (06 Marks)
- b. Apply merge sort to sort the following numbers in ascending order: 8, 3, 2, 9, 7, 1, 5, 4. Also obtain time complexity for merge sort. (08 Marks)
- c. Apply divide and conquer approach to multiply the following matrices:

$$\begin{bmatrix} 5 & 2 & 6 & 1 \\ 0 & 6 & 2 & 0 \\ 3 & 8 & 1 & 4 \\ 1 & 8 & 5 & 6 \end{bmatrix} \times \begin{bmatrix} 7 & 5 & 8 & 0 \\ 1 & 8 & 2 & 6 \\ 9 & 4 & 3 & 8 \\ 5 & 3 & 7 & 9 \end{bmatrix}$$

(06 Marks)

OR

- 4 a. Write an iterative algorithm for binary search and trace the algorithm to search for a key 151 in a list -15, -6, 0, 7, 9, 23, 54, 82, 101, 112, 125, 131, 142, 151. (06 Marks)
- b. Write quicksort algorithm to sort 'n' numbers and apply the same to sort the following numbers in ascending order. 80, 60, 70, 40, 10, 30, 50, 20. (08 Marks)

- c. Apply DFS and source vertex removal methods to obtain topological sequence for two graph shown in Fig.Q.4(c)
(06 Marks)

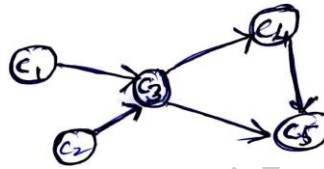


Fig.Q.4(c)

Module-3

- 5 a. Apply greedy method to obtain an optimal task assignment for the 7 tasks and infinite supply of machines given their start end times.

Task	a	b	c	d	e	f	g
Start time	0	3	4	9	7	1	6
End time	2	7	7	11	10	5	8

(06 Marks)

- b. Solve the following knapsack problem using all 3 greedy criteria: $m = 20$, $n = 3$, $p = (25, 24, 15)$ $w = (18, 15, 10)$.
(06 Marks)
- c. Write and apply kruskals algorithm to obtain minimum cost spanning tree for the graph shown in Fig.Q.5(c).
(08 Marks)

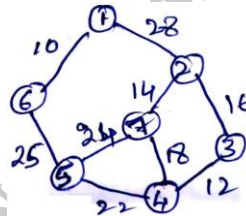


Fig.Q.5(c)

OR

- 6 a. Write and apply prims algorithm to obtain minimum cost spanning tree for the graph shown in Fig.Q.5(c).
(08 Marks)
- b. Apply Dijkstra algorithm to find the shortest distance from vertex 1 to all nodes in a graph shown in Fig.Q.6(b)
(06 Marks)

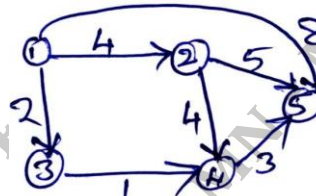


Fig.Q.6(b)

- c. Determine the optimal prefix code for the symbols a, o, q, u, y, z that occur with frequencies: 20, 28, 4, 17, 12, 7 respectively.
(06 Marks)

Module-4

- 7 a. Define multistage graph problem and write the forward approach algorithm to obtain a solution.
(06 Marks)
- b. Write Floyd's algorithm to solve all pairs shortest path problem and apply the same for the graph shown in Fig.Q.7(b).
(08 Marks)

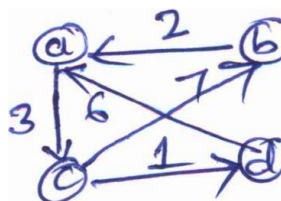


Fig.Q.7(b)

- c. Find optimal tour for the travelling sales person problem using dynamic programming technique for the graph shown in Fig.Q.7(c) considering initial and end vertex as 1.

(06 Marks)

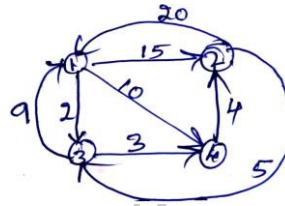


Fig.Q.7(c)

OR

- 8 a. Define transitive closure of a digraph. Find the transitive closure matrix for the graph whose

adjacency matrix is
$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 1 & 0 & 1 & 0 \end{bmatrix}$$

(08 Marks)

- b. Write the algorithm to construct optimal binary search tree for the following data:

Key	A	B	C	D
Probability	0.1	0.2	0.4	0.3

(12 Marks)

Module-5

- 9 a. Define n-queens problem. Construct state-space tree for solving 4-queens problem using backtracking for all possible solutions. (08 Marks)
- b. Solve the following assignment problem using branch and bound technique, whose cost matrix for assigning four jobs to four persons are given.

$$\begin{bmatrix} 9 & 2 & 7 & 8 \\ 6 & 4 & 3 & 7 \\ 5 & 8 & 1 & 8 \\ 7 & 6 & 9 & 4 \end{bmatrix}$$

(06 Marks)

- c. Explain the classes of NP-hard and NP-complete. (06 Marks)

OR

- 10 a. State the subset sum problem. Using backtracking, obtain a solution to the subset sum problem given $s = \{6, 8, 2, 14\}$ and $d = 16$. (06 Marks)
- b. Construct a space tree representing all possible colorings using atmost 3 colors for the graph shown in Fig.Q.10(b). (07 Marks)

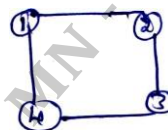


Fig.Q.10(b)

- c. With the help of a state-space tree, solve the travelling sales person problem Fig.Q.10(c) using branch-and-bound technique. (07 Marks)

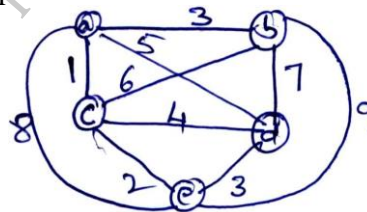


Fig.Q.10(c)

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Fourth Semester B.E. Degree Examination, June/July 2024
Data Communication

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define Data Communication. Explain fundamental characteristics of data communication system. (06 Marks)
- b. State five basic data communication components, briefly explain their functions. (06 Marks)
- c. What are the advantages of having layered model for networking? Explain important functions of OSI model. (08 Marks)

OR

- 2 a. What is Transmission Impairment? Briefly explain three causes of transmission impairment. (06 Marks)
- b. Explain briefly about Shannon capacity and Nyquist bit rate for communication channels. (06 Marks)
- c. Distinguish :
 - i) Baseband and Broadband
 - ii) Bandwidth and Throughput
 - iii) Latency and Jitter
 - iv) Dejure and Defacto model. (08 Marks)

Module-2

- 3 a. Define Line Coding. List out its characteristics. (08 Marks)
- b. Explain different data transmission modes in detail. (07 Marks)
- c. Represent the sequence 1011001011 using polar and biphase schemes. (05 Marks)

OR

- 4 a. Explain with suitable diagram PCM encoder used for analog to digital conversion. (08 Marks)
- b. Briefly explain with neat diagrams, ASK and FSK modulation techniques and specify the bandwidth requirement. (08 Marks)
- c. We have an available bandwidth of 100 kHz which spans from 200 to 300 kHz. What are the carrier frequency and the bit rate if we modulated our data by using ASK with $d = 1$. (04 Marks)

Module-3

- 5 a. What is Switching? What are the different types of switching techniques? (06 Marks)
- b. What is Hamming distance? Discuss about minimum Hamming distance. (06 Marks)
- c. Explain the need for multiplexing. Four 1kbps connections are multiplexed together. A unit is 1 bit find :
 - i) The duration of 1 bit before multiplexing
 - ii) The transmission rate of the link
 - iii) The duration of a time slot
 - iv) The duration of a frame. (08 Marks)

OR

- 6 a. Describe about Direct Sequence Spread Spectrum (DSSS). (06 Marks)
 b. Distinguish statistical TDM from synchronous TDM. (06 Marks)
 c. Explain Cycle Redundancy Check (CRC). Assume that data is 10110 and the code generator is 1101. Calculate CRC bits. (08 Marks)

Module-4

- 7 a. Demonstrate taking an example, character oriented and bit oriented framing. (10 Marks)
 b. A network transmit 200bit frames on a shared 200 Kbps line. Compute the throughput for pure ALOHA and slotted ALOHA if the system produces
 i) 1000 frames/sec ii) 500 frames/sec iii) 250 frames/sec.
 Tabulate the values computed. (10 Marks)

OR

- 8 a. Demonstrate the concept of IP address and Link – layer address, consider a small internet. (07 Marks)
 b. What is the role of Address Resolution Protocol (ARP)? Explain its Operation. (07 Marks)
 c. What is Classless Inter Domain Routing (CIDR)? Explain Address Aggregation Strategy with example. (06 Marks)

Module-5

- 9 a. Explain Ethernet frame format with a neat diagram. (08 Marks)
 b. Describe Gigabit Ethernet. (06 Marks)
 c. Explain the architecture of IEEE 802.11. (06 Marks)

OR

- 10 a. Explain the architecture of Bluetooth. (08 Marks)
 b. Explain the operation of cellular telephony. (06 Marks)
 c. Explain fourth generation (4G) of cellular telephony. (06 Marks)

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CBCS SCHEME

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18CS51

Fifth Semester B.E. Degree Examination, June/July 2024 Management and Entrepreneurship for IT Industry

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain management and discuss the functional areas of management. (08 Marks)
- b. Explain the steps involved in planning. (06 Marks)
- c. Discuss the staff selection process in detail (06 Marks)

OR

- 2 a. Bring out the nature of management and discuss the various roles of a manager. (08 Marks)
- b. List the types of organizations and explain any two highlighting its advantages and disadvantages. (06 Marks)
- c. Compare and contrast strategic and tactical planning. (06 Marks)

Module-2

- 3 a. Explain different leadership styles with their merits and demerits. (08 Marks)
- b. Discuss the characteristics and importance of coordination. (06 Marks)
- c. Discuss Hertzberg's two factors theory. (06 Marks)

OR

- 4 a. Explain motivation. Discuss Maslow's need hierarchy theory of motivation. (08 Marks)
- b. Explain controlling and discuss the steps in controlling. (06 Marks)
- c. Explain the importance of communication. (06 Marks)

Module-3

- 5 a. Discuss the different types of entrepreneurs in detail. (10 Marks)
- b. Explain financial and technical feasibility study. (10 Marks)

OR

- 6 a. Explain the stages of entrepreneurial process in detail. (10 Marks)
- b. Discuss the role of entrepreneurs in the economic development of the country and list the barriers to entrepreneurship. (10 Marks)

Module-4

- 7 a. Explain the significance of project report and list down the planning commission guidelines for project report preparation. (08 Marks)
- b. Illustrate the functional areas of operation of management – Marketing and sales and supply chain management. (06 Marks)
- c. Write a note on project identification with sources. (06 Marks)

OR

- 8 a. Discuss the various contents of project report. (08 Marks)
- b. Illustrate the functional areas of operation of management – Accounting and finance and human resources. (06 Marks)
- c. Write a note on project selection. (06 Marks)

Module-5

- 9 a. Discuss the steps involved in establishing micro and small enterprise. (08 Marks)
b. Discuss the case study of Infosys (N R Narayana Murthy). (06 Marks)
c. Explain the objectives and functions provided by SIDBI and KIADB (06 Marks)

OR

- 10 a. Explain the objectives and functions provided by
(i) KSFC (ii) DIC (iii) TECSOC (iv) KSSIDC (08 Marks)
b. Discuss the case study of Air Deccan (Captain G. R. Gopinath). (06 Marks)
c. Explain Trademark, Copy Rights and Patents. (06 Marks)

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CBCS SCHEME

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18CS52

Fifth Semester B.E. Degree Examination, June/July 2024 Computer Networks and Security

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Differentiate between non-persistent and persistent connections in HTTP. (05 Marks)
- b. Explain the conditional GET operation. (05 Marks)
- c. Illustrate file distribution time in peer to peer and client server architecture. (10 Marks)

OR

- 2 a. Explain mail transfer from sender to receiver using SMTP protocol. (10 Marks)
- b. Explain DNS Records and Messages in detail. (10 Marks)

Module-2

- 3 a. With a neat diagram, explain TCP segment structure. (07 Marks)
- b. Explain the causes and costs of congestion. (08 Marks)
- c. Elaborate the three way handshake in TCP. (05 Marks)

OR

- 4 a. Explain network assisted congestion control in ATM Available Bit Rate (ABR). (06 Marks)
- b. Explain reliable data transfer in a channel with bit errors. (06 Marks)
- c. In detail explain the selective repeat protocol for reliable data transfer. (08 Marks)

Module-3

- 5 a. What is routing? With a neat diagram, explain the structure of a router. (10 Marks)
- b. Explain link state routing algorithm. Compute the shortest path for the network shown in Fig.Q5(b) using link state algorithm.

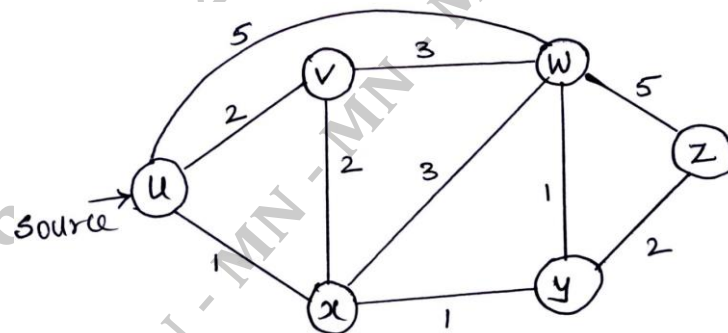


Fig.Q5(b)

(10 Marks)

OR

- 6 a. Explain IPv6 packet format in detail. (07 Marks)
- b. Explain the significance of spanning tree in broadcast routing. (05 Marks)
- c. Explain inter-AS routing in the internet with BGP protocol. (08 Marks)

Module-4

- 7 a. Explain the threats to network security. (08 Marks)
b. Explain RSA algorithm. Using RSA encrypt a message $m = 9$. Assume $p = 3$, $q = 11$ and $x = 3$. Compute y and show encryption and decryption. (08 Marks)
c. Explain encryption in advanced encryption standard. (04 Marks)

OR

- 8 a. In the Diffie Hellman key exchange protocol, prove that the two keys K_1 and K_2 are equal. (05 Marks)
b. With a neat diagram, discuss the steps in DES algorithm. (10 Marks)
c. Write a note on firewalls. (05 Marks)

Module-5

- 9 a. Explain multimedia streaming using HTTP. (08 Marks)
b. What are the properties of video? (04 Marks)
c. Discuss loss anticipation schemes used by VOIP applications. (08 Marks)

OR

- 10 a. Briefly discuss how DNS redirects a user request to a CDN server with an example. (08 Marks)
b. Explain setting up a call to a known IP address in SIP. (08 Marks)
c. Explain RTP packet header. (04 Marks)

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Fifth Semester B.E. Degree Examination, June/July 2024 Database Management Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Discuss the main characteristics of database approach over file-processing approach. (10 Marks)
- b. Explain the operations of 2-Tier and 3-Tier client/server architecture of DBMS. (10 Marks)

OR

- 2 a. What is a weak entity type? Explain the role of partial key in design of weak entity type. (05 Marks)
 - b. Design an ER diagram for the mail order database considering the following requirements.
Employee takes order for parts from customers:
 - i) Employees are identified by unique employee ID, first name and last name, address, gender, zip code.
 - ii) Customer is identified by a unique customer ID, first and last name, address, location, zip code.
 - iii) Part is identified by a unique part number, part name, price and quantity.
 - iv) Order is identified by a unique order number, date of receipt, expected ship date, actual ship date. Each order contains specified quantities of one or more parts.
 - v) Each customer can place number of orders and each order is placed by one customer only.
 - vi) Each employee can take any number of orders but each order belongs to only one employee.
 - vii) Each part is placed by number of customers and each customer can place order for number of parts.
- Write assumptions made. (10 Marks)
- c. Differentiate specialization and generalization, giving suitable examples. (05 Marks)

Module-2

- 3 a. List and explain the different characteristics of relations. (08 Marks)
- b. With an example, discuss the basic constraints that can be specified when you create a table in SQL. (06 Marks)
- c. Write queries in relational algebra for the following: [Refer tables given in question 5(b)].
 - i) Retrieve the number of dependents for an employee named "Ram".
 - ii) Retrieve the name of managers working in location named "XYZ" who has no female dependents.
 - iii) Retrieve the name of employee who works in the same department as that of "Raj". (06 Marks)

OR

- 4 a. Briefly discuss the different types of update operations on relational database. Give examples for the violation of referential integrity in each of the update operation. (10 Marks)
- b. With examples, explain the steps of ER to relational mapping algorithm. (10 Marks)

Module-3

- 5 a. What is a view in SQL? Explain with examples. Discuss the problems that may arise when one attempts to update a view. (10 Marks)
- b. Consider the following tables:
 Employee (Name, Ssn, Salary, Superssn, Dno)
 Department (Dname, Dno, Mgrssn, Mgrstartdate)
 Project (Pname, Pno, Plocation, Dno)
 Dept_Location (DNum, Dlocation)
 Works_on (Essn, Pnum, Hours)
 Dependent (Essn, Depname, Sex)
- i) List the names of managers who have at least one dependent.
- ii) For each employee, retrieve the employee's name and name of his or her immediate supervisor.
- iii) For each project on which more than two employees work, retrieve the project number, project name and the number of employees who work on that project.
- iv) Retrieve the name of employees whose salary is greater than salary of all the employees working in either department 5 or 6. (10 Marks)

OR

- 6 a. What is a cursor in embedded SQL? Explain with examples. (10 Marks)
- b. With examples, explain the following:
- i) Java Script
- ii) Style sheets. (10 Marks)

Module-4

- 7 a. List and explain the informal design guidelines for relation schema. (10 Marks)
- b. What are prime and non-prime attributes? Explain with examples. (04 Marks)
- c. Consider the relation $R = \{A, B, C, D, E, F, G, H, I, J\}$ and the set of functional dependencies (FDs) $F = \{AB \rightarrow C, BD \rightarrow EF, AD \rightarrow GH, A \rightarrow I, H \rightarrow J\}$. What is the key of R? Decompose R into ZNF and 3NF relations. (06 Marks)

OR

- 8 a. Consider the two sets of FD's:
 $F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$ and $G = \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$. Show that they are equivalent. (06 Marks)
- b. Consider a relation $R(A, B, C, D)$ with $FDS = \{A \rightarrow BC, B \rightarrow C, A \rightarrow B, AB \rightarrow C\}$. Find the minimal cover for the set of FDs. (06 Marks)
- c. Write and explain the algorithm for dependency-preserving and non additive join decomposition into 3NF schemes with suitable example. (08 Marks)

Module-5

- 9 a. What is serializability? Explain serial, non serial and conflict-serializable schedules with appropriate examples. (10 Marks)
- b. Discuss the time stamp ordering algorithm for concurrency control. How does strict time stamp ordering differ from basic time stamp ordering? (10 Marks)

OR

- 10 a. What is a Deadlock? Consider the following sequences of actions listed in the order they are submitted to DBMS sequence $S1 : R1(A), W2(B), R1(B), R3(C), W2(C), W4(B), W3(A)$. Draw waits for graph in case of deadlock situation. (06 Marks)
- b. Explain shadow paging with suitable example. (06 Marks)
- c. Briefly explain the recovery techniques based on deferred update and immediate update. (08 Marks)

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Fifth Semester B.E. Degree Examination, June/July 2024 Automata Theory and Computability

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following terms with examples:
 i) Alphabet ii) String iii) Language iv) Power of alphabet v) Σ^* (08 Marks)
- b. Design DFSM for the following languages:
 i) $L = \{W \text{ in } \{a, b\}^* : \text{string } W \text{ end with } abb\}$
 ii) $L = \{W \text{ in } \{0, 1\}^* : \text{string } W \text{ being with } 01\}$
 iii) Set of all strings of 0's and 1's with substring 110 (12 Marks)

OR

- 2 a. i) Convert the following NDFSM to equivalent DFSM. [Refer Fig.Q2(a)]

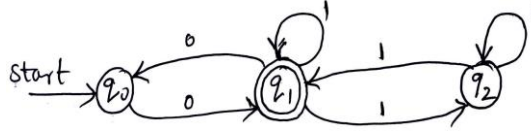


Fig.Q2(a)

(05 Marks)

- ii) Construct DFSM from the following ϵ -NDFSM.

δ	ϵ	a	b	c
$\rightarrow p$	{ q, r }	ϕ	{ q }	{ r }
q	ϕ	{ p }	{ r }	{ p, q }
*r	ϕ	ϕ	ϕ	ϕ

(05 Marks)

- b. Define Equivalent and Distinguishable pair of states. Construct minimum state DFSM for the following DFSM.

δ	a	B
$\rightarrow 1$	2	4
* 2	3	6
3	2	4
* 4	6	5
5	2	4
6	6	6

(10 Marks)

Module-2

- 3 a. Define Regular Expression. Design Regular Expression for the following Languages.
 i) $L = \{a^m b^n : (m + n) \text{ is even}\}$
 ii) $L = \{a^m b^n : m \geq 4, n \leq 3\}$
 iii) Set of all strings of 0's and 1's with atleast one occurrence of 00 (08 Marks)
- b. Prove that Regular Grammar define exactly Regular Language. (06 Marks)
- c. Convert the following Regular expressions to equivalent FSM.
 (i) $(a + b)^* ab$ (ii) $(aa)^* + (bb)^*$ (06 Marks)

OR

- 4 a. State and prove pumping theorem for Regular Languages. (08 Marks)
 b. Show that $L = \{a^n b^n : n \geq 1\}$ is not Regular Language. (06 Marks)
 c. Define Regular Grammar. Design Regular Grammar for the following Languages:
 i) $L = \{W \text{ in } \{a, b\}^* : |W| \text{ is even}\}$
 ii) Set of all strings of a's and b's which end with ab (06 Marks)

Module-3

- 5 a. Design Context Free Grammar for the following languages:
 (i) Set of all strings of a's and b's with equal number of each.
 (ii) $L = \{a^i b^j c^k : k = i + j\}$
 (iii) $L = \{a^{2m} b^n : m \geq 1, n \geq 1\}$
 (iv) $L = \{a^n b^n c^n : n \geq 1\}$ (10 Marks)
 b. Construct (i) left Most Derivation (ii) Right Most Derivation (iii) Parse tree for the string $W = aaabab$ using the grammar.
 $S \rightarrow AbB \quad A \rightarrow aA \mid \epsilon \quad B \rightarrow aB \mid bB \mid \epsilon$ (10 Marks)

OR

- 6 a. Define PDA. Design PDA for the following language.
 $L = \{W \text{ in } \{a, b\}^* : n_a(W) = n_b(W)\}$
 Number of a's is same as number of b's
 Write Transition diagram of PDA and instantaneous description of PDA for the input string $W = abba$. (14 Marks)
 b. Define CNF. Convert the following grammar to CNF.
 $S \rightarrow ABa \mid a$
 $A \rightarrow aab \mid b$
 $B \rightarrow Ac \mid c$ (06 Marks)

Module-4

- 7 a. Define Turing Machine. Design Turing Machine for $L = \{a^n b^n : n \geq 1\}$
 b. Write transition diagram of T.M and also write sequence of ID's of T.M for the input string $W = aabb$. (14 Marks)
 c. Explain the model of Linear Bounded Automata with a diagram. (06 Marks)

OR

- 8 a. Explain different techniques of Turing Machine Construction. (10 Marks)
 b. Explain Multitape Turing Machine with a diagram. (06 Marks)
 c. Explain Non-Deterministic Turing Machine. (04 Marks)

Module-5

- 9 a. Explain Post Correspondence Problem. (07 Marks)
 b. Explain Halting problem of Turing Machine. (07 Marks)
 c. Explain Decidability and Decidable languages. (06 Marks)

OR

- 10 a. Explain Quantum Computers. (07 Marks)
 b. Explain Church – Turing Thesis (06 Marks)
 c. Explain Class P and Class NP (07 Marks)

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Fifth Semester B.E. Degree Examination, June/July 2024 Application Development using Python

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. List the salient features of python programming language. (06 Marks)
- b. Write a python program to find the area of sphere and volume of cube. Print the results. Take input from user. (06 Marks)
- c. List and explain the syntax of all flow control statements with example. (08 Marks)

OR

- 2 a. What is a function? How to define a function in python? Write a program using function to find out the given number is prime or not. (06 Marks)
- b. What is local and global scope of variable in python? Explain the different scenarios with an example snippet. (08 Marks)
- c. What is Exception Handling? How exceptions are handled in python? Write a python program with exception handling code to solve divide-by-zero error situation. (06 Marks)

Module-2

- 3 a. What is list? Explain the concept of slicing and indexing with proper examples. (06 Marks)
- b. What is Tuple? How it is different from list? Write a program to count the number of occurrences of character in a string. (06 Marks)
- c. What is Dictionary in Python? How it is different from list and tuples? Write a program to create, update and display the dictionary items. (08 Marks)

OR

- 4 a. List out five useful string methods. Explain with an example code for each method. (10 Marks)
- b. Compare copy.copy() and copy.deepcopy() functions with suitable examples for each. (05 Marks)
- c. Write a python program that accepts a sentence and find the number of words, digits, characters, uppercase letters and lowercase letters. (05 Marks)

Module-3

- 5 a. List and explain shorthand code for common character classes. Illustrate how do you define your own character class. (07 Marks)
- b. Explain the usage of caret and dollar sign characters in regular expression. (06 Marks)
- c. Write a python program to extract phone numbers and email addresses using regex. (07 Marks)

OR

- 6 a. What are the Ray properties of a file? Explain in detail file reading/writing process with an example of python program. (07 Marks)
- b. Explain briefly what are the different methods of file operations support in python shutil module. (07 Marks)
- c. Write a python program to create a folder PYTHON and under the hierarchy 3 files file1, file2 and file3. Write the content in file1 as "XXX" and in file 2 as "YYY" and file 3 content should be by opening and merge of file1 and file2. Check out necessary condition before writing file3. (06 Marks)

Module-4

- 7 a. Define classes and objects in python. Create a class called employee and initialize it with employee id and name. Design methods to:
- i) SetAge – to assign age to employee.
 - ii) SetSalary – to assign salary to the employee.
 - iii) Display – to display all information of the employee. (08 Marks)
- b. Illustrate the concept of modifier with python code. (05 Marks)
- c. Explain `__init__` and `__str__` method with an example python program. (07 Marks)

OR

- 8 a. Define Polymorphism. Demonstrate polymorphism with function to find histogram to count the number of times each letter appears in a word and in a sentence. (07 Marks)
- b. Illustrate the concept of pure function with python code. (06 Marks)
- c. Define class diagram. Discuss the need for representing class relationships using class diagram with suitable example. (07 Marks)

Module-5

- 9 a. How do we download a file and save it to hard drive using request module? (06 Marks)
- b. Write a python program to give search keyword from command line arguments and open the browser tab for each result page. (06 Marks)
- c. Explain Selenium's web drive methods for finding elements. (08 Marks)

OR

- 10 a. Write a program that takes a number N from command line and creates an $N \times 100$ multiplication table in excel spread sheet. (08 Marks)
- b. Write short notes on:
Creating, copying and rotating pages with respect to pdf. (06 Marks)
- c. Write a program that find all CSV files in the current working directory, read in the full contents of each file, write out the contents, skipping the first line, to a new CSV file. (06 Marks)

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18CS81

Eighth Semester B.E. Degree Examination, June/July 2024 Internet of Things

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define IoT and discuss the genesis of IoT in detail. (06 Marks)
- b. List out the difference between IT and OT networks and their various challenges. (06 Marks)
- c. What are the different challenges of IOT? Explain. (08 Marks)

OR

- 2 a. Explain with diagram the one M2M IoT standardized architecture. (08 Marks)
- b. Explain IoT Data Management and Compute Stack. (08 Marks)
- c. List and explain the defining characteristics of fog computing. (04 Marks)

Module-2

- 3 a. List out the most useful classification scheme for the pragmatic application of sensors in a IoT network. (08 Marks)
- b. Define sensors and actuators. Explain how they interact with the physical world. (08 Marks)
- c. Define smart objects. Explain its characteristics. (04 Marks)

OR

- 4 a. What are constrained devices and constrained node networks? Classify them. (08 Marks)
- b. Explain Zigbee protocol stack using IEEE 802.15.4. (08 Marks)
- c. Briefly describe about communication criteria. (04 Marks)

Module-3

- 5 a. What are the key advantages of the IP suite for IoT? (08 Marks)
- b. Explain in detail the 6LOWPAN. (08 Marks)
- c. Explain the different schedule management and packet forwarding models of TiSCH. (04 Marks)

OR

- 6 a. Explain in detail COAP message format. (08 Marks)
- b. Explain Message Queuing Telemetry Transport (MQTT). (06 Marks)
- c. Explain the raw socket tunneling of SCADA using different scenarios. (06 Marks)

Module-4

- 7 a. What are the ways IoT data is categorized? Explain in detail. (08 Marks)
- b. Explain in detail supervised learning and unsupervised learning. (06 Marks)
- c. Explain in detail the core functions of edge analytics with necessary diagrams. (06 Marks)

OR

- 8 a. Explain the different steps and phases of OCTAVE Allegro methodology. (08 Marks)
- b. Explain Lambda Architecture in details. (06 Marks)
- c. Explain any two Big data Analytics tools and technologies. (06 Marks)

Module-5

- 9 a. What is Arduino? What are the advantages of Arduino? (08 Marks)
b. How to install arduino software for the windows PCs? (06 Marks)
c. Explain the different pins/parts of Arduino Uno Board. (06 Marks)

OR

- 10 a. Explain the different layers of IoT smart layered architecture. (08 Marks)
b. Explain smart parking architecture with advantages and disadvantages. (06 Marks)
c. With a neat diagram, explain wireless temperature monitoring system using Raspberry Pi. (06 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2024 Data Mining and Data Warehousing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is Data Warehouse? Explain three tier architecture of data warehouse with a neat diagram. (10 Marks)
- b. Explain the various differences between OLTP and OLAP systems. (10 Marks)

OR

- 2 a. Explain with suitable examples the various OLAP operations in a multi-dimensional data model. (10 Marks)
- b. Explain the concept of star, snowflake and fact constellation schemas for multi-dimensional databases with examples. (10 Marks)

Module-2

- 3 a. What is data mining? With a neat diagram, explain KDD process in data mining. Also explain the various data mining tasks with suitable examples. (12 Marks)
- b. For the following vectors X and Y, calculate
 - (i) Cosine similarity
 - (ii) Euclidean distance
 - (iii) Jaccard coefficient
 - (iv) Correlation coefficient
$$X = (0, 1, 0, 1); Y = (1, 0, 1, 0)$$
(08 Marks)

OR

- 4 a. With respect to indexing, explain bit map index and join index operations. (10 Marks)
- b. Briefly explain any five data preprocessing approaches. (10 Marks)

Module-3

- 5 a. What is frequent itemset generation? Explain frequent itemset generation using Apriori principle with an example. (10 Marks)
- b. Describe alternative methods for generating frequent itemsets. (10 Marks)

OR

- 6 a. Briefly explain the factors affecting the computational complexity of apriori algorithm. (04 Marks)
- b. What is association analysis? Explain association rule, support and confidence with an example. (06 Marks)
- c. Explain objective measures of interestingness for evaluating association patterns. (10 Marks)

Module-4

- 7 a. Define classification. With a neat diagram explain the general approach to solve classification problem. (08 Marks)
- b. Illustrate Hunt's algorithm to develop a decision tree. Using hunt's algorithm derive decision tree for the following data :

	Binary	Categorical	Continuous	Class
Tid	Home owner	Marital status	Annual Income	Defaulted Borrower
1	Yes	Single	125 K	No
2	No	Married	100 K	No
3	No	Single	70 K	No
4	Yes	Married	120 K	No
5	No	Divorced	95 K	Yes
6	No	Married	60 K	No
7	Yes	Divorced	220 K	No
8	No	Single	85 K	Yes
9	No	Married	75 K	No
10	No	Single	90 K	Yes

(12 Marks)

OR

- 8 a. What is a rule based classifier? Explain the following :
 (i) Sequential covering algorithm.
 (ii) Rule ordering schemes. (10 Marks)
- b. Write an algorithm for K-nearest neighbour classification. List the characteristics of nearest neighbor classifiers. (10 Marks)

Module-5

- 9 a. What is Cluster analysis? Explain the different types of clustering techniques with examples. (10 Marks)
- b. Explain K-means clustering algorithm. What are its limitations? (10 Marks)

OR

- 10 a. Explain Agglomerative hierarchical clustering algorithm with different proximity between clusters. (10 Marks)
- b. Explain DBSCAN algorithm and estimate time and space complexity. How the parameters are selected? (10 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2024 Object Oriented Modelling and Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following terms with example and UML notation: Multiplicity, link and Association, Qualified association, Association class and Enumeration. (10 Marks)
- b. What is multiple inheritance? Explain the kinds of multiple inheritance. (10 Marks)

OR

- 2 a. Briefly explain aggregation and composition. Give their respective UML notations with an example. (10 Marks)
- b. Explain association end and N-ary association with example. (10 Marks)

Module-2

- 3 a. Explain the detailed notation and alternate notation to show the repeating operation in SSD's with example. (10 Marks)
- b. What is the purpose of state chart? Explain the primary steps for developing a state chart. (10 Marks)

OR

- 4 a. What are use cases and actors? Explain use case diagram with an example. (10 Marks)
- b. Explain state diagram for class "OrderItem" with an example. (10 Marks)

Module-3

- 5 a. Explain the steps involved in constructing a domain class model. (10 Marks)
- b. Briefly explain stages of software development. (10 Marks)

OR

- 6 a. What is system conception? Explain the ways to devise and elaborate a system concept. (10 Marks)
- b. Explain the steps to construct the domain state model. (10 Marks)

Module-4

- 7 a. Explain the process of design with communication diagrams. (10 Marks)
- b. Explain design classes and design within class diagrams. (10 Marks)

OR

- 8 a. Explain fundamental design principles to object oriented design. (10 Marks)
- b. List and explain primary responsibility of three layers design. (10 Marks)

Module-5

- 9 a. Explain how to select and how to use a design pattern. (10 Marks)
- b. What is design pattern? List and explain essential elements of design pattern. (10 Marks)

OR

- 10 a. Write a note on :
 - (i) Proxy design pattern
 - (ii) Singleton design pattern. (10 Marks)
- b. Explain the process of how design pattern solve design problem. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg. 42+8 = 50, will be treated as malpractice.

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18CS734

Seventh Semester B.E. Degree Examination, June/July 2024

User Interface Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define User Interface Design. Explain the need of user interface design in user's perspective. (08 Marks)
- b. Explain the importance of user interface design. (04 Marks)
- c. Discuss Direct and Indirect manipulation. (08 Marks)

OR

- 2 a. Explain general principles of user interface design. (08 Marks)
- b. Explain the important characteristics of graphical user interface. (08 Marks)
- c. Differentiate between graphical and web user interface. (04 Marks)

Module-2

- 3 a. Explain 5 commandments in designing for people. (07 Marks)
- b. List and explain the pitfalls in development path of the design process. (07 Marks)
- c. Discuss the necessity of human interaction speed. (06 Marks)

OR

- 4 a. Explain in detail business definition and requirement analysis in user's perspective. (10 Marks)
- b. Explain the importance of obstacles and usability partial objectives. (10 Marks)

Module-3

- 5 a. Explain in detail functions of menus with example. (10 Marks)
- b. Determine the basic functionalities of navigating menus and graphical menus. (10 Marks)

OR

- 6 a. Explain the structures of menus in detail. (10 Marks)
- b. Discuss Website Navigation in detail. (10 Marks)

Module-4

- 7 a. Define windows. Explain the important characteristics of windows. (10 Marks)
- b. Discuss windows presentation styles. (10 Marks)

OR

- 8 a. List and explain types of windows. (08 Marks)
- b. Explain window operations in detail. (06 Marks)
- c. Explain the important characteristics of device-based controls. (06 Marks)

Module-5

- 9 a. Discuss screen based controls. (10 Marks)
- b. Write a note on different prototypes used in user interface design. (10 Marks)

OR

- 10 a. Compare user test and usability test. Explain the process involved in developing and conducting a test. (10 Marks)
- b. Write a short note on the following :
 - i) Radio buttons
 - ii) Check boxes
 - iii) List boxes
 - iv) Drop-down/pop-up list boxes
 - v) Palettes(10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
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Seventh Semester B.E. Degree Examination, June/July 2024

Digital Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Explain the components of a general purpose image processing system with diagram. (10 Marks)
- b. Explain the concept of sampling and quantization with diagrams. (10 Marks)

OR

- 2 a. Explain the fundamental steps in digital image processing. (10 Marks)
- b. Define 4-adjacency, 8-adjacency and m-adjacency. (06 Marks)
- c. If P and Q are two pixels at coordinates (100, 120) and (130, 160) respectively. Find chess board and cityblock distance. (04 Marks)

Module-2

- 3 a. Explain Price – Wise – Linear – Transformation functions. (10 Marks)
- b. What are the applications of arithmetic and logical operations? (10 Marks)

OR

- 4 a. Discuss smoothing linear filters and order statistics filter. (10 Marks)
- b. Describe how first order derivatives are used for image enhancement. (10 Marks)

Module-3

- 5 a. Discuss one dimensional Discrete Fourier Transform and its inverse. (10 Marks)
- b. Explain translation and separability properties of DFT. (10 Marks)

OR

- 6 a. What are the basic steps for filtering in frequency domain? Explain with a neat diagram. (08 Marks)
- b. Explain the following frequency domain filter
Ideal Lowpass filter, Butterworth low pass filter, Gaussian lowpass filter. (12 Marks)

Module-4

- 7 a. What are the three basic types of gray level discontinuities? Explain point detection. (10 Marks)
- b. Describe region splitting and merging with example. (10 Marks)

OR

- 8 a. Define image segmentation and explain line detection. (10 Marks)
- b. Explain basic global thresholding. (10 Marks)

Module-5

- 9 a. What are the principles types of data redundancies? Discuss. (10 Marks)
- b. Explain the general image compression system with necessary diagram. (10 Marks)

OR

- 10 a. Write short note on : i) Huffman coding ii) LZW coding. (10 Marks)
- b. What is image compression? Where is it used? List the differences between lossy and lossless compression. (10 Marks)

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18CS823

Eighth Semester B.E. Degree Examination, June/July 2024

NOSQL Database

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Enlist the features of NOSQL database. (04 Marks)
- b. Summarize features of column family models. (08 Marks)
- c. Discuss advantages and disadvantages of schemalessness (08 Marks)

OR

- 2 a. Explain factors contributing to the emergency of NOSQL databases. (08 Marks)
- b. Explain Impedance mismatch with the help of suitable example. (07 Marks)
- c. Describe Materialized Views. (05 Marks)

Module-2

- 3 a. What is a role of quorum in maintaining the consistency? (06 Marks)
- b. Explain CAP theorem. (04 Marks)
- c. What is version stamp? What are the ways to create version stamps? (10 Marks)

OR

- 4 a. Explain single server, master slave and peer to peer distribution models. (12 Marks)
- b. With the help of suitable example explain write-write consistency and read-write consistency. (08 Marks)

Module-3

- 5 a. Consider the example of product sales data in each year. Describe MapReduce process to compare the sales of product for each month in 2011 to the prior year. (08 Marks)
- b. Identify the situations where i) Key-value store is ideal ii) Key-value store is not best solution. (06 Marks)
- c. Write short note on increment MapReduce process. (06 Marks)

OR

- 6 a. Explain the importance of partitioning and combining in the MapReduce process. (08 Marks)
- b. Explain key value store features with respect to consistency, transactions, query features, structure of the data and scaling. (12 Marks)

Module-4

- 7 a. Compare features of Oracle with MongoDB. (04 Marks)
- b. How to ensure consistency with availability in MongoDB? (12 Marks)
- c. Why are document stores not suitable for transactions or queries with varying aggregate structures? (04 Marks)

OR

- 8 a. Describe scaling and sharding in MongoDB. (08 Marks)
- b. Explain schema changes and incremental migration in MongoDB. (08 Marks)

- c. i) Write query in SQL and MongoDB to select orderId and orderDate for a customer with ID as IN_BL_12182. (02 Marks)
- ii) Write equivalent query in MongoDB for a given SQL query.
 select * FROM customerorder, orderItem, product
 where
 customerOrder.orderID = orderItem.customerOrderID
 AND orderItem.productID = product.productID
 AND product.name LIKE '%Refactoring%' (02 Marks)

Module-5

- 9 a. Describe the procedure to add indexing for the nodes in the NE04J database. (08 Marks)
- b. Consider Barbara is connected to Jill by two distinct paths; write query with explanation to
 i) Find all these paths and the distance between Barbara and Jill along those different paths.
 ii) Find shortest path between Barbara and Jill using pathfinder and Dijkstra's algorithm. (12 Marks)

OR

- 10 a. Explain Graph database. List out use cases of Graph database. (08 Marks)
- b. Write cipher queries to
 i) Find all outgoing relationships with the type of Friend and return the friends names of person "AAAAAA" for greater depth than one.
 ii) Find relationships where a particular relationship property exists. Filter on the properties of relationships and query if a property exists or not. (06 Marks)
- c. Describe the ways to scale in graph databases. (06 Marks)

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Third Semester B.E. Degree Examination, June/July 2024 Data Structures and Applications

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Write a program in C to demonstrate how whole structure is passed as a parameter to a function. (04 Marks)
- b. Define DMA. List and explain different DMA functions used in C. (08 Marks)
- c. Explain the representation of linear array in memory and give example. (08 Marks)

OR

- 2 a. Consider two polynomials $A(x) = 7x^{1000} + 4$ and $B(x) = x^4 + 5x^3 + 4x^2 + 3$. Show diagrammatically how these two polynomials can be stored in a single dimensional array. (04 Marks)
- b. Define polynomial and degree of the polynomial. Write the representation of polynomial using array and structures. (08 Marks)
- c. Write a program in C to read sparse matrix of integer values and to search the sparse matrix for an element specified by the user. (08 Marks)

Module-2

- 3 a. Define Stack. Give the C implementation of push and pop function. Include check for empty and full condition of stack. (07 Marks)
- b. Convert the following infix expression into prefix and postfix expressions:
i) $((H * (((A + ((B + C) * D)) * F) * G) * E)) + J$
ii) $A/B - C + D * E - A * C$ (08 Marks)
- c. Write a program in c to implement tower of Hanoi using recursive function. (05 Marks)

OR

- 4 a. Write a function in c to add, delete and display the elements from queue. (07 Marks)
- b. Write a program in c to implement the operations on a circular queue using dynamically allocated arrays. (08 Marks)
- c. What is priority queue? Briefly explain the types of priority queues. (05 Marks)

Module-3

- 5 a. Write a program in C to implement Stack operations using single linked list. (07 Marks)
- b. Write a program in C to implement Queue operations using single linked list. (08 Marks)
- c. Write a program in C to count the number of nodes in a single linked list. (05 Marks)

OR

- 6 a. Write a program in C to implement insert front, delete front and display functions using double linked list. (07 Marks)

- b. Write a linked representation for the given sparse matrix.

$$A = \begin{bmatrix} 2 & 0 & 0 & 3 \\ 0 & 0 & 5 & 0 \\ 0 & -1 & 0 & 0 \\ 0 & 0 & 0 & 8 \\ 0 & 7 & 4 & 0 \end{bmatrix}$$

Fig.Q.6(b)

- c. Differentiate between single linked list and double linked list.

(08 Marks)

(05 Marks)

Module-4

- 7 a. Define binary tree and state its properties. Show how binary tree is represented using an array and linked list. (08 Marks)
- b. Write the binary tree for the expression $A/B * C * D + E$. Write the result of preorder and post order traversals for the given expression. (07 Marks)
- c. Write the algorithm for preorder and post order traversals. (05 Marks)

OR

- 8 a. Define Threaded Binary Tree. Write the memory representation of Threaded Binary Tree for the given graph. (08 Marks)

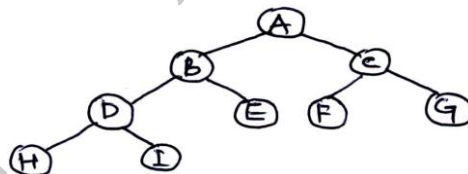


Fig.Q.8(a)

- b. Draw the binary search tree for the following inputs and write recursive function to search for a given key value.

13 4 7 3 17 21 15 19 2 23

(07 Marks)

- c. Write the applications of trees.

(05 Marks)

Module-5

- 9 a. Define graph. What are the different methods of representing a graph? Give example. (10 Marks)
- b. Define the following with an example:
- Directed graph
 - Multigraph
 - Complete graph
 - Cyclic and acyclic graph
 - Loop.

(10 Marks)

OR

- 10 a. Define BFS with an example. Write a function in C to implement BFS. (10 Marks)
- b. What is Hashing? Briefly explain the different types of hashing techniques. Construct the hash table for storing C built-in functions, acos, define, float, exp, char, atan, ceil, floor
Note: Use hash table with 26 buckets and 2 slots per bucket. (10 Marks)

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Third Semester B.E. Degree Examination, June/July 2024
Analog and Digital Electronics

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. With a neat circuit diagram and mathematical analyses explain voltage divider bias circuit. (10 Marks)
- b. With a neat circuit diagram explain the working of relaxation oscillator. (10 Marks)

OR

- 2 a. List the advantages of active filters over passive filters. (05 Marks)
- b. Explain any two performance parameters of power supply. (05 Marks)
- c. Explain R-2R ladder type D to A converter. (10 Marks)

Module-2

- 3 a. Simplify the following expression using K-Map and draw the logic circuit using basic gates.
 $F(A, B, C, D) = \sum m(7) + \sum d(10, 11, 12, 13, 14, 15)$. (10 Marks)
- b. Simplify the expression using Quine-McClusky method $F(A, B, C, D) = \sum m(0, 2, 3, 6, 7, 8, 10, 12, 13)$. (10 Marks)

OR

- 4 a. Using K-Map obtain SOP and POS expressions for
 $f(A, B, C, D) = \sum m(6, 8, 9, 10, 11, 12, 13, 14, 15)$. (10 Marks)
- b. With example explain,
 - i) Map-Entered variable method
 - ii) Petricks method. (10 Marks)

Module-3

- 5 a. What is Hazard? With example explain static-0 and static-1 Hazards. (10 Marks)
- b. Implement following function using 8:1 multiplexer:
 $f(a, b, c, d) = \sum m(0, 1, 5, 6, 8, 10, 12, 15)$. (10 Marks)

OR

- 6 a. Implement full adder using 3:8 decoder and NAND gates. (10 Marks)
- b. Design 7-segment decoder using PLA. (10 Marks)

Module-4

- 7 a. What are the 3 modeling styles in VHDL? Write VHDL code for full adder using structural model. (10 Marks)
- b. Derive the characteristics equations for D, T, SR and JK flip flops. (10 Marks)

OR

- 8 a. Draw the logic diagram of master slave JK flipflop using NAND gates and explain its working. (10 Marks)
- b. Explain the working of SR flipflop and show how it can be used in debounce circuit. (10 Marks)

Module-5

- 9 a. What is shift register? What are the different types of shift registers? Explain 8 bit serial-in serial-out shift register. (10 Marks)
- b. Explain the following:
- i) Sequential Parity Checker
 - ii) Ring Counter. (10 Marks)

OR

- 10 a. Design Mod-5 synchronous counter using JK flip flop. (10 Marks)
- b. With diagram explain parallel adder with accumulator. (10 Marks)

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21CS42

Fourth Semester B.E. Degree Examination, June/July 2024 Design and Analysis of Algorithms

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define algorithm. Explain the following asymptotic notations:
i) Big On ii) Big Omega iii) Big Theta (08 Marks)
b. Design a non-recursive algorithm to find maximum element in an array of n elements. Give the Mathematical Analysis. (08 Marks)
c. Define time and space complexity. (04 Marks)

OR

- 2 a. Design an algorithm for performing sequential search and compute Best, Worst and Average case efficiency. (08 Marks)
b. Write an algorithm to find the uniqueness of element an array and give mathematical analysis of this non recursive algorithm with steps. (08 Marks)
c. List and explain basic asymptotic efficiency classes. (04 Marks)

Module-2

- 3 a. Illustrate the tracing of the Quick Sort algorithm for the following set of numbers
5, 3, 1, 9, 8, 2, 4, 7 (10 Marks)
b. Apply the topological sorting algorithm for the following graph shown in Fig.Q3(b). Find the topological sequence.

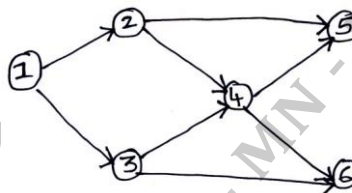


Fig.Q3(b)

(10 Marks)

OR

- 4 a. Write a C++/JAVA program for Merge Sort. Analyze its efficiency and apply the same to sort the following numbers : 4, 9, 0, -1, 6, 8, 9, 2, 3, 12. (10 Marks)
b. Write a recursive algorithm for Binary Search and also bring out its efficiency. (10 Marks)

Module-3

- 5 a. Apply greedy method to obtain an optimal solution to the knapsack problem given M = 60.
W = {5, 10, 20, 30, 40}
P = {30, 20, 100, 90, 160}
Find total profit earned. (10 Marks)

- b. Apply single source shortest path algorithm to the following graph Fig.Q5(b). Assume vertex 'a' as source.

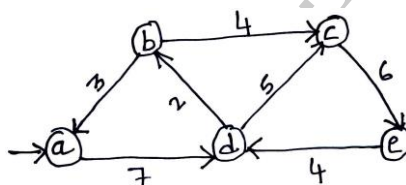


Fig.Q5(b)

(10 Marks)

OR

- 6 a. A message consisting of the character given in the table below has to be transmitted network in a secured manner.

Character	A	M	R	-
Probability	0.4	0.2	0.3	0.1

- Construct Huffman tree
- Device Huffman codes for the given characters
- Encode the text : RAMA RAMAR
- Decode the text : 1000101

(10 Marks)

- b. Find the optimal solution using greedy for the job sequencing with dead line problem with following values:

Job	J ₁	J ₂	J ₃	J ₄	J ₅
Profit	10	3	33	11	40
Dead line	3	1	1	2	2

(10 Marks)

Module-4

- 7 a. Define a Multistage Graph. Give an example. Explain the technique of finding the minimum cost path in a multistage graph. (10 Marks)
- b. Write Floyd's Algorithm and find all pair Shortest path for the given graph. [Refer Fig.Q7(b)]

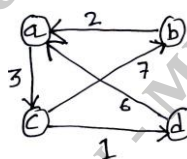


Fig.Q7(b)

(10 Marks)

OR

- 8 a. Apply the Dynamic Programming to solve travelling sales person problem for the following graph shown in Fig.Q8(a).

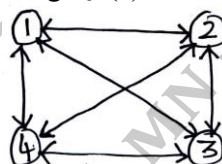


Fig.Q8(a)

C	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0

(10 Marks)

- b. Write Horspool Algorithm for string matching. Trace the algorithm to find the pattern "ELECTION" in the text.

"EDUCATION ONLY HELPS IN SELECTION."

(10 Marks)

Module-5

- 9 a. Construct the state-space tree for sum of subset problem given the following data:
 $W = \{3, 5, 6, 7\}$ and $m = 15$. (10 Marks)
- b. Write C++ / JAVA program to find all Hamiltonian cycles in a connected undirected Graph G of n vertices using backtracking principle. (10 Marks)

OR

- 10 a. Explain Branch and Bound concept. Apply Branch and Bound to the following instance of assignment problem.

	Job1	Job2	Job3	Job4
Person A	9	2	7	8
Person B	6	4	3	7
Person C	5	8	1	8
Person D	7	6	9	4

- b. Explain the following concepts : (10 Marks)
- i) Graph coloring problem with an example
 - ii) NP Complete Problem
 - iii) NP-Hard Class Problem (10 Marks)

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CBCS SCHEME

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21CS44

Fourth Semester B.E. Degree Examination, June/July 2024

Operating Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Compare multiprogramming and Time sharing systems. (06 Marks)
- b. Illustrate with a diagram the Dual mode operation of an operating system. (08 Marks)
- c. Point out and explain briefly the various services provided by the operating systems. (06 Marks)

OR

- 2 a. What are system calls? Explain types of system calls. (06 Marks)
- b. What are microkernels? Point out its benefits. (06 Marks)
- c. Define Process. What are its states? Explain process state diagram. (08 Marks)

Module-2

- 3 a. What is multithreading? Explain different types of multithreading models. (08 Marks)
- b. For the following example calculate average waiting time and average turnaround time for the following algorithms.

i) FCFS ii) Preemptive SJF iii) Round Robin (Time quantum = 1 ms)

Process	Arrival time	Burst time
P ₁	0	8
P ₂	1	4
P ₃	2	9
P ₄	3	5

(12 Marks)

OR

- 4 a. Explain Critical Section Problem. What are its requirements? (08 Marks)
- b. Explain Dining Philosophers problem. Illustrate the solution to Dining philosopher problem using semaphores. (12 Marks)

Module-3

- 5 a. What is Deadlock? What are its necessary conditions for the occurrence of deadlock? (08 Marks)
- b. Consider the following snapshot of a system.

Process	Allocation				MAX				Available			
	A	B	C	D	A	B	C	D	A	B	C	D
P ₀	0	0	1	2	0	0	1	2	1	5	2	0
P ₁	1	0	0	0	1	7	5	0				
P ₂	1	3	5	4	2	3	5	6				
P ₃	0	6	3	2	0	6	5	2				
P ₄	0	0	1	4	0	6	5	6				

Answer the following question using the Bankers Algorithm.

- i) What is the content of Need matrix?
- ii) Is the system is in Safe state?
- iii) If the request from process P₁ arrives for (0, 4, 2, 0) can the request be granted immediately? (12 Marks)

OR

- 6 a. Given memory partitions of 100K, 500K, 200K, 300K and 600K (in order) how would each of the first fit, best fit and worst fit algorithms place processes of 212K, 417K, 112K and 426K (in order) which algorithm makes the most efficient use of memory. (06 Marks)
- b. What is paging? With a neat diagram explain Paging hardware. (08 Marks)
- c. Differentiate between Segmentation and Paging. (06 Marks)

Module-4

- 7 a. What do you mean by Page Replacement? Explain the working of page replacement algorithm with a neat block diagram. (08 Marks)
- b. Consider the following page-reference string
2, 3, 2, 1, 5, 2, 4, 5, 3, 2, 5, 2
How many page faults occur in the following replacement algorithms, assuming three frames:
i) FIFO ii) LRU iii) Optimal (12 Marks)

OR

- 8 a. Describe the various file allocation methods. Also point out their advantages and disadvantages. (10 Marks)
- b. Explain the various Free Space Management techniques. (10 Marks)

Module-5

- 9 a. What is disk scheduling? Explain different types of Scheduling algorithms. (10 Marks)
- b. What is access matrix? How the access matrix is implemented, point out the advantages and disadvantages of different methods of implementation of access matrix. (10 Marks)

OR

- 10 a. Explain the different components of a Linux system. (10 Marks)
- b. Discuss the Linux file system. (10 Marks)

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CBCS SCHEME

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21CS51

Fifth Semester B.E. Degree Examination, June/July 2024 Automata Theory and Compiler Design

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define the following terminologies with appropriate examples and notations:
 - i) Kleene star ii) Alphabet iii) Language iv) Power of an alphabet (08 Marks)
- b. Design a DFA to accept the language
 $L = \{ w \mid w \text{ is of even length and begins with } 01 \}$ (07 Marks)
- c. Explain briefly phases of a compiler. (05 Marks)

OR

- 2 a. Define with terminologies different ways of representing Automata considering an example. (08 Marks)
- b. Consider the following ϵ -NFA Fig.Q2(b).

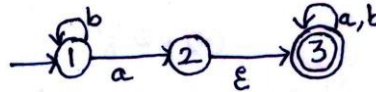


Fig.Q2(b)

- i) Compute the ϵ -Closure of each state ii) Convert the automation to a DFA (07 Marks)
- c. Explain in brief commonly used compiler construction tools. (05 Marks)

Module-2

- 3 a. Write regular expression (RE) for the following Languages.
 - i) The set of all strings such that the number of 0's is ODD. $\Sigma = \{0, 1\}$
 - ii) Every ODD length string begins with 11. $\Sigma = \{0, 1\}$ (08 Marks)
- b. Convert the following FSM into RE using state elimination technique. Refer Table Q3(b).

δ	0	1
$\rightarrow q_1$	q_2	q_1
q_2	q_2	q_4
$*q_3$	q_4	q_2
$*q_4$	q_4	q_1

Table Q3(b)

- c. Describe the languages denoted by the following regular expressions :
 - i) $a.(a + b)^*.b$ ii) $(a + b)^*.a.(a + b)(a + b)$ (05 Marks)

OR

- 4 a. Write Regular Expressions for the following languages:
 - i) All strings of lowercase letters that contain the five vowels in order.
 - ii) All the strings of a's and b's that contain the substring abb. (08 Marks)
- b. Convert the following DFA in Fig.Q4(b) to a Regular Expression using Kleene's theorem.

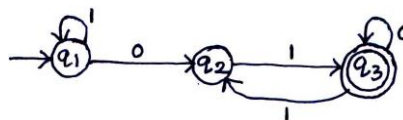


Fig.Q4(b)

- c. Explain with neat diagram interactions between the lexical analyzer and the parser. (05 Marks)

Module-3

- 5 a. Design Context-Free Grammars (CFG) for the following languages:
- $L = \{a^i b^j c^k \mid i = j = k\}$
 - The set of all strings of 0's and 1's where the number of 0's is equal to the number of 1's. (08 Marks)
- b. Given the Context-Free Grammar below:
- $$S \rightarrow AS \mid \varepsilon$$
- $$A \rightarrow aa \mid ab \mid ba \mid bb$$
- Give leftmost and rightmost derivations and parse tree for the following strings:
- aaba
 - baabab
 - aaabbb
- (06 Marks)
- c. Construct the top-down parse tree for string $w = id + id * id$ by using grammar given below:
- $$E \rightarrow TE'$$
- $$E' \rightarrow +TE' \mid \varepsilon$$
- $$T \rightarrow FT'$$
- $$T' \rightarrow *FT' \mid \varepsilon$$
- $$F \rightarrow (E) \mid id$$
- (06 Marks)

OR

- 6 a. Remove ambiguity from the Grammar given below:
- $$S \rightarrow aSb$$
- $$S \rightarrow aaSb$$
- $$S \rightarrow \varepsilon$$
- (08 Marks)
- b. Consider the Context – Free Grammar given below:
- $$S \rightarrow aB \mid bA$$
- $$A \rightarrow a \mid aS \mid bAA$$
- $$B \rightarrow b \mid bS \mid aBB$$
- The string $w = 'aaabbabbba'$ and find
- Left-most derivation
 - Right-most derivation
 - Parse-tree
- (06 Marks)
- c. Explain the role of Parser in the compiler model. (06 Marks)

Module-4

- 7 a. Design PDA to accept the language
- $$L = \{WcW^R \mid W \in \{a, b\}^*\}$$
- Write ID for $W = 'bacab'$ (10 Marks)
- b. Construct bottom-up parse tree for the following input strings by considering grammar given below:
- $$E \rightarrow E + T \mid T$$
- $$T \rightarrow T * F \mid F$$
- $$F \rightarrow (E) \mid id$$
- $W_1 = id * id$ $W_2 = id + id * id$ (10 Marks)

OR

- 8 a. Design a ND-PDA to accept the language
- $$L = \{a^m b^n \mid m \neq n, n, m > 0\}$$
- and write ID for $W = aaabb$ (10 Marks)
- b. Explain LR – Parsing algorithm in detail. (10 Marks)

Module-5

- 9 a. Design Turing Machine for the language
 $L = \{ a^i b^i \mid i > 0 \}$
Write ID for string $W = "aabb"$ (10 Marks)
- b. Write a short note on the following topics:
i) Recursive Languages ii) Universal Turing Machines (10 Marks)

OR

- 10 a. Construct Direct Acyclic Graph (DAG) and corresponding three address code for the following expressions:
i) $a + a * (b - c) + (b - c) * d$
ii) $((x + y) - ((x + y) * (x - y))) + ((x + y) * (x - y))$ (10 Marks)
- b. Write a short note on the following :
i) Multitape Turing Machine
ii) Non-Deterministic Turing Machine (10 Marks)

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Fifth Semester B.E. Degree Examination, June/July 2024

Database Management Systems

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define DBMS. Explain main characteristics of database approach. (08 Marks)
- b. Explain advantages of Database Management system. (08 Marks)
- c. Define data model. Explain the different types of user friendly interfaces. (04 Marks)

OR

- 2 a. Define Entity and attributes. Explain all the types of attributes along with notations. (10 Marks)
- b. Explain Cardinality ratios for binary relationship and write a ER diagram for movie database (minimum 4 entities). (10 Marks)

Module-2

- 3 a. Explain relational model constraints. (06 Marks)
- b. Explain different types of update operations and show an example of a violation of the referential and entity integrity in each of update operation. (08 Marks)
- c. Define the following with example : (06 Marks)
 - (i) Primary key
 - (ii) Foreign key
 - (iii) Super key
 - (iv) Candidate key

OR

- 4 a. Briefly explain the ER to relational mapping algorithm with suitable example for each step. (10 Marks)
- b. Explain following relational algebra operators with example : (10 Marks)
 - (i) Select
 - (ii) Project
 - (iii) Intersection
 - (iv) Cartesian product

Module-3

- 5 a. Explain insert, delete, update, alter and drop statement in SQL. (10 Marks)
- b. Consider the following schema for order database : (10 Marks)

SALESMAN (Salesman_Id, Name, City, Commision)

CUSTOMER (Customer_Id, Cust_name, City, Grade, Salesman_id)

ORDERS (Ord_No, Purchase_amt, Ord_Date, Customer_id, Salesman_id)

Write SQL queries to,

 - (i) Find the name and numbers of all salesman who had more than due customer.
 - (ii) List all the salesman and indicates those who have and don't have customer in their cities (use union).
 - (iii) Create that view finds the salesman who has the customers with the higher order.

OR

- 6 a. Write a note on for following :
 (i) Assertion and action trigger. (10 Marks)
 (ii) Views in SQL. (05 Marks)
 b. Explain stored procedures in SQL. (05 Marks)
 c. Briefly explain JDBC classes. (05 Marks)

Module-4

- 7 a. Explain informal guidelines to determine the quality of relation scheme design with example. (08 Marks)
 b. Explain Armstrong inference rule. (06 Marks)
 c. Discuss insertion and deletion anomalies. (06 Marks)

OR

- 8 a. Define normal form. Explain 2NF, 3NF and BCNF with suitable example. (10 Marks)
 b. Consider 2 sets of FDs, F and G, $F = \{A \rightarrow B, B \rightarrow C, AC \rightarrow D\}$ and $G = \{A \rightarrow B, B \rightarrow C, A \rightarrow D\}$ Are F and G equivalent? (05 Marks)
 c. Consider set of FD's be $E : \{B \rightarrow A, D \rightarrow A, AB \rightarrow D\}$, find the minimal cover of E. (05 Marks)

Module-5

- 9 a. Why concurrency control needed. Explain types of problems that may occur when 2 simple transaction run concurrently. (10 Marks)
 b. Explain why recovery needed and Acid properties. (10 Marks)

OR

- 10 a. Briefly discuss Two-phase locking techniques for concurrency control. (08 Marks)
 b. Explain ARIES recovery algorithm with example. (08 Marks)
 c. Write a note on Deadlock prevention protocol. (04 Marks)

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Sixth Semester B.E. Degree Examination, June/July 2024 Software Engineering and Project Management

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. Define software process. Explain generic software process framework. (10 Marks)
- b. Define process patterns. Explain the means of describing the patterns. (10 Marks)

OR

- 2 a. Explain the different types of evolutionary process models. (10 Marks)
- b. Explain waterfall model. (10 Marks)

Module-2

- 3 a. Define requirements engineering. Explain its distinct tasks. (10 Marks)
- b. Explain various approaches in requirements modeling. (10 Marks)

OR

- 4 a. Explain requirements elicitation. (10 Marks)
- b. Explain preliminary use case diagram for the Safe Home system. (10 Marks)

Module-3

- 5 a. Explain principles of agility. (10 Marks)
- b. Explain the process of extreme programming. (10 Marks)

OR

- 6 a. Explain scrum process model. (10 Marks)
- b. Explain Feature Driven Development (FDD). (10 Marks)

Module-4

- 7 a. Explain the significance of efficient project management. (10 Marks)
- b. Define project. Explain the characteristics of a project. (10 Marks)

OR

- 8 a. Explain the different ways of categorizing software projects. (10 Marks)
- b. Explain the activities of management in doing management control. (10 Marks)

Module-5

- 9 a. Define software quality. Explain quality specification in detail. (10 Marks)
- b. Why do we need software quality models? Explain Garvin's quality dimension. (10 Marks)

OR

- 10 a. Explain McCall's model. (10 Marks)
- b. Explain ISO 9126's major external software quality characteristics. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg. 42+8 = 50, will be treated as malpractice.

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Sixth Semester B.E. Degree Examination, June/July 2024 Full Stack Development

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is web framework? Explain django-admin command with example. (06 Marks)
- b. Explain the history of django. (07 Marks)
- c. Explain view function in django with an example. (07 Marks)

OR

- 2 a. Illustrate how django processes request. (08 Marks)
- b. Identify the key philosophy behind URL confs and loose coupling. (06 Marks)
- c. Describe the process of mapping URLs to views. (06 Marks)

Module-2

- 3 a. What is template? Explain basics of template systems with example. (10 Marks)
- b. Explain template inheritance. (10 Marks)

OR

- 4 a. Identify the different types of tags and filters in djang template system. (12 Marks)
- b. Explain the models in django with example. (08 Marks)

Module-3

- 5 a. How to activate and configure the admin interface in django for managing application data? (08 Marks)
- b. Explain the process of handling and processing forms in a web application using django. (12 Marks)

OR

- 6 a. Develop a django program to create feedback forms. (08 Marks)
- b. Discuss the usage of the admin interface. (06 Marks)
- c. How to create form in django? What this class can do in python interpreter? (06 Marks)

Module-4

- 7 a. Discuss the concept of generic views of objects in django. (10 Marks)
- b. Explain cookies with example. (10 Marks)

OR

- 8 a. How does user authentication works in django? (10 Marks)
- b. Discuss the role of site map frame work. (10 Marks)

Module-5

- 9 a. Explain technologies on which AJAX overlaid. (10 Marks)
- b. Discuss about jQuery and basic Ajax. (10 Marks)

OR

- 10 a. Write a note on java script and XMIHHP response. (10 Marks)
- b. Illustrate the following : i) CSS ii) JSON iii) HTML iv) Iframe (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg. 42+8 = 50, will be treated as malpractice.

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Sixth Semester B.E. Degree Examination, June/July 2024 Computer Graphics and Fundamentals of Image Processing

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What is computer graphics illustrate the application of computer graphics. (10 Marks)
- b. Illustrate display window management system using GLUT. (10 Marks)

OR

- 2 a. Using Bresenham's line drawing algorithm digitize the line with end points (20, 10) to (30, 18). (10 Marks)
- b. With a simple OpenGL program demonstrate the different OpenGL functions. (10 Marks)

Module-2

- 3 a. Illustrate the need of homogeneous co-ordinate system and demonstrate translation, rotation, scaling in 2D homogeneous co-ordinate system with matrix representation. (10 Marks)
- b. Obtain a matrix representation for rotation and scaling of a object about a specified pivot point in 2D. (10 Marks)

OR

- 4 a. Illustrate the raster method for geometric transformation. (10 Marks)
- b. List and explain all 3-D geometric transformation. (10 Marks)

Module-3

- 5 a. Illustrate the logical classification of input devices. (08 Marks)
- b. Elaborate the following with the suitable OpenGL function using code snippet: (12 Marks)
 - i) GLUT mouse function
 - ii) GLUT keyboard function.

OR

- 6 a. Demonstrate the steps in design of animation. (08 Marks)
- b. Illustrate the use of morphing with edge equalization and vertex equalization. (12 Marks)

Module-4

- 7 a. What is image processing? List some of the fields of Ip. (04 Marks)
- b. List the types of images based on nature, attribute and colour. (06 Marks)
- c. Let $V = \{0, 1\}$, compute the D_e , D_4 , D_8 and D_m distance between 2 pixel p and q. Let the pixel coordinates of p and q be (3, 0) and (2, 3). (10 Marks)

0 1 1 1
 1 0 0 1
 1 1 1 1(q)
 1 1 1 1
 (p)

OR

- 8 a. Describe image interpolation technique. (10 Marks)
b. List and explain arithmetic operations by considering an example. (10 Marks)

Module-5

- 9 a. What is image segmentation? Describe the types of segmentation algorithm. (10 Marks)
b. With the help of flow chart. Explain the stages of edge detection. (10 Marks)

OR

- 10 a. Write a brief note on:
i) Canny edge detection (10 Marks)
ii) Graph theoretic algorithm (10 Marks)
b. Explain the basic type of grey level discontinuities in a digital image. (10 Marks)

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Question Paper Version : A

Fourth Semester B.E. Degree Examination, June/July 2024**Unix Shell Programming**

Time: 1 hr.]

[Max. Marks: 50

INSTRUCTIONS TO THE CANDIDATES

1. Answer all the **fifty** questions, each question carries one mark.
2. Use only **Black ball point pen** for writing / darkening the circles.
3. **For each question, after selecting your answer, darken the appropriate circle corresponding to the same question number on the OMR sheet.**
4. Darkening two circles for the same question makes the answer invalid.
5. **Damaging/overwriting, using whiteners** on the **OMR** sheets are strictly prohibited.

-
1. Which of the following are command oriented operating systems :
(i) DOS (ii) Windows (iii) UNIX (iv) Linux
a) (i) and (ii) b) (i) and (iii)
c) (iii) and (iv) d) (i) and (iv)
 2. UNIX System is invented by
a) Ken Thompson and Denni's Ritchie b) Richard Stallman and Linus Torvalds
c) Sumitabha Das d) Kumar
 3. The UNIX System is written in
a) C++ b) C c) java d) Python
 4. Date command in Unix
a) Display only date
b) Display both date and time
c) Display data, time and calendar of the month
d) Set the date and time
 5. On seeing the symbol, the shell performs the job of connecting the two commands.
a) | b) > c) + d) <
 6. _____ is the command used to clear the screen in Unix system.
a) clrscr b) clear c) tty d) tput clear
 7. Choose the correct statement:
a) The shell interact with the machine hardware and the kernel with the user
b) Both shell and kernel interact with machine hardware
c) The kernel interact with the machine hardware and the shell with the user
d) None of the above

8. Which program loads the kernel into memory at start up?
a) Shell program b) System call c) Operating system d) Bootstrap program
9. Choose correct pair to fill the blanks :
_____ have places and _____ have life.
a) Maps, earth b) Files, process
c) Kernel, shell d) Operating system, Application software
10. A program starts to run at terminal, as soon as you login is _____
a) shell b) kernel c) tty d) calendar
11. Following is the File Type in Unix:
a) Ordinary Files b) Directory Files
c) Device Files d) All of these
12. While naming UNIX files which of the following is an incorrect statement?
a) File names may or may not have extensions
b) You are not permitted to use control characters or other unprintable characters
c) File names are case sensitive
d) File name can have as many dots embedded in its name.
13. When you log in to the system Unix automatically places you in a directory called
a) bin b) kumar c) home d) root
14. Which command is used for checking current working directory?
a) ls b) dir c) pwd d) cwd
15. When cd command is invoked without arguments
a) It reverts to home directory. b) It display the current directory
c) It generates error message d) No action taken
16. System refuses to create a directory due to
a) The directory already exists
b) There is a ordinary file with same name in current directory
c) Don't have permissions to create a file or directory
d) All of these
17. In Unix . (dot) is not used
a) representing the current directory b) To name hidden files
c) To separate extension part of file name d) To represent a single character wild card
18. Identify the file that is not made available during system installation
a) /dev b) /var c) /etc d) /bin
19. Identify the correct statement with respect to Unix files.
a) The file is a container for storing information
b) Unix files does not contain eof (end-of-file) marker
c) A file's name and size are not stored in the file
d) All of these

20. match ls command options to corresponding descriptions:
- | | |
|-------|--|
| (1) R | (5) Sorts file names by last access time |
| (2) r | (6) Shows all filenames with a dot (.) |
| (3) a | (7) Multi columnar output |
| (4) u | (8) Recursive list |
| | (9) Sorts filenames in reverse order |
- a) 1 – 8 , 2 – 9 , 3 – 6 , 4 – 5
 b) 1 – 6 , 2 – 7 , 3 – 9 , 4 – 5
 c) 1 – 7 , 2 – 9 , 3 – 5 , 4 – 6
 d) 1 – 7 , 2 – 5 , 3 – 8 , 4 – 9
21. Command used for listing file attributes is
 a) list – l b) ls – l c) ls – d d) list – a
22. Both UID and GID are maintained in
 a) /etc/group b) /user c) /etc/passwd d) /dev
23. A link count greater than one in file attributes indicates that
 a) The file has more than one name
 b) There are more copies of the file
 c) There are more owners of the file
 d) There are more processes using the file
24. Command used to change the group ownership
 a) chown b) chgroup c) chgrp d) changegrpowner
25. Which statement is true with respect to hardlinks in Unix systems?
 a) You can not have two linked filenames in two file systems.
 b) You can not link a directory even within the same file system.
 c) Both a and b
 d) Neither a nor b
26. Choose the correct permission set after executing the Unix command \$chmod 666 x start
 a) – rw – rw – rw – b) – rw – r – x – – x
 c) – rwxrwxrwx d) – r – xr – xr – x
27. When user create files, the Unix system has the following default permissions for all files except dictionaries.
 a) – rw – rw – rw – b) – rw – r – – r – –
 c) – rwxrwxrwx d) – – – – – – – –
28. Who set the privileges of the group with respect to a file?
 a) group owner b) File owner
 c) Super user d) Operating system
29. File's permissions can only be changed by
 a) Owner of the file b) Group member
 c) System administrator d) Both a and c

30. \$chmod 000 xstart
 a) Above command is invalid, No error reported by shell
 b) It is valid command, But shell neglects this, hence no change in existing permission set for the file xstart
 c) It is invalid command, Error is reported by shell
 d) It is valid command, resulting in permission set changes to _____ for the file xstart
31. A shell program runs in _____ mode.
 a) interface b) super user c) interpretive d) interactive
32. #! In a shell script , represents
 a) command line b) Positional parameter c) interpreter line d) Nothing
33. Match the following :
- | A | B |
|-------------------------|--|
| Special Shell Parameter | Significance |
| (1) \$* | (5) Exit status of last command |
| (2) \$# | (6) Complete set of positional parameters as a single string |
| (3) \$? | (7) PID of the last background job |
| (4) \$! | (8) Number of arguments specified in command line |
- a) 1 – 6 , 2 – 8 , 3 – 5 , 4 – 7 b) 1 – 8 , 2 – 6 , 3 – 7 , 4 – 5
 c) 1 – 5 , 2 – 7 , 3 – 6 , 4 – 8 d) 1 – 7 , 2 – 5 , 3 – 8 , 4 – 6
34. test command cannot be used for :
 a) comparing two numbers
 b) comparing two strings or a single one for a null value
 c) checking a file's attributes
 d) searching for a pattern in a file
35. case statement in shell handles string tests, more efficiently than if :
 a) Above statement is False.
 b) Above statement is True.
 c) Case statement is not existing in shell
 d) Case statement cannot compare two strings
36. What is the output of following commands:
 \$x = 5
 \$x = 'expr \$x+1'
 \$ echo \$x
 a) 6 b) \$x c) Error d) 5
37. The shell featured loops are :
 (1) while (2) until (3) for
 a) Only (1) b) Only (3)
 c) Both (1) and (3) d) (1), (2) and (3)

38. Apart from assigning values to positional parameters, set serves as a useful debugging tool with the option:
 a) – d b) – debug c) – x d) – xd
39. _____ is used in the same script as the command to provide nominal standard input.
 a) Read document b) Here document c) From document d) get document
40. Shell's signal handling command is
 a) signal b) catch c) trap d) except
41. Process is
 a) An application b) A copy of stored program
 c) A instance of a running program d) None of these
42. When you log on to a Unix system, a process that immediately setup by the kernel is
 a) Shell process b) Bootstrap process
 c) init process d) User process
43. ps command is used to
 a) Print a document b) Display process status
 c) Create a new process d) Modify process attributes
44. The system calls associated with mechanism of process creation :
 (1) fork (2) exec (3) Wait
 a) Only (1) b) Only (1) and (2) c) (1), (2) and (3) d) None of these
45. _____ command is used to schedule jobs to run repeatedly.
 a) time b) cron c) batch d) at
46. Identify the correct statement:
 a) A job is the name given to a group of processes
 b) Way to creating a job is to run a pipeline of two or more commands
 c) You can use shell job control facilities to manipulate jobs
 d) All of these
47. Identify a command that permits execution of the process after the user has logged out.
 a) nohup b) wait
 c) ps d) No such command exists
48. nice command with the & operator used to
 a) Reduce the priority of the jobs b) Increase the priority of the jobs
 c) Set equal priority to all jobs d) None of these
49. Premature death of the Parent process makes the Child process _____
 a) An orphan process b) to Die
 c) A zombie process d) init process
50. Processes whose parents don't wait for their death move to the _____
 a) runnable state b) sleeping state
 c) ghost state d) zombie state

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CBCS SCHEME

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21CS642

Sixth Semester B.E. Degree Examination, June/July 2024

Advanced Java Programming

Time: 3 hrs.

Max. Marks: 100

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-1

- 1 a. What are enumerations? Explain compareTo(), equals() and values() method with example. (10 Marks)
- b. Explain how to obtain Annotation at Run Time by using Reflection. (10 Marks)

OR

- 2 a. Explain the following with examples: (i) wrappers (ii) autoboxing (iii) unboxing (10 Marks)
- b. Define annotations. Explain the following built in annotations: (i) override (ii) inherited (10 Marks)

Module-2

- 3 a. Define Generics. Explain the generic class with two type parameters. (10 Marks)
- b. Write a program to illustrate bounded wildcards with respect to generics. (10 Marks)

OR

- 4 a. Elaborate how raw type is created in generic class and list its disadvantages. (10 Marks)
- b. Describe the ambiguity of errors in generic class. (10 Marks)

Module-3

- 5 a. Define string. Write a program to demonstrate the use of constructors in string. (10 Marks)
- b. Explain how to modify a string by using following methods:
(i) Substring() (ii) Replace() (iii) Trim() (10 Marks)

OR

- 6 a. Illustrate any four character extraction method with examples. (10 Marks)
- b. What is StringBuffer? Explain any five StringBuffer methods with example. (10 Marks)

Module-4

- 7 a. Define cookie. List out methods defined by cookie. Write a Java program to add a cookie. (10 Marks)
- b. Write a Servlet program that read parameter names and values of print. (10 Marks)

OR

- 8 a. Define JSP. Explain different types of JSP tags with suitable examples. (10 Marks)
- b. Develop a Servlet program that illustrate how to use cookies. (10 Marks)

Module-5

- 9 a. List and explain JDBC Drivers types. (10 Marks)
- b. List and explain the various statements objects in JDBC. (10 Marks)

OR

- 10 a. Explain the three kinds of exceptions occurred in JDBC. (10 Marks)
- b. Explain transaction processing in JDBC. Write a program to execute a Database transaction. (10 Marks)

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Important Note : 1. On completing your answers, compulsorily draw diagonal cross lines on the remaining blank pages.
2. Any revealing of identification, appeal to evaluator and /or equations written eg, 42+8 = 50, will be treated as malpractice.

CBCS SCHEME

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BBOC407

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024 Biology for Engineers (CSE)

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Discuss the various components of Eukaryotic cells.	10	L3	CO1
	b.	Identify the applications of stem cells.	5	L2	CO1
	c.	Explain the functions of vitamins.	5	L2	CO1
OR					
Q.2	a.	Compare Prokaryotic and Eukaryotic cells.	10	L3	CO1
	b.	Explain the properties of Carbohydrates.	5	L2	CO1
	c.	Explain the functions of Lipids.	5	L2	CO1
Module – 2					
Q.3	a.	Highlighting the properties of cellulose, justify cellulose as an effective water filter.	10	L3	CO1
	b.	Explain the working and development of DNA vaccines by taking suitable example.	10	L2	CO1
OR					
Q.4	a.	What are Bioplastics? Justify the use of PHA as Bioplastic mentioning its properties and applications.	10	L3	CO1
	b.	Discuss the following : (i) Meat analogs of protein. (ii) Lipids as cleaning agents.	10	L2	CO1
Module – 3					
Q.5	a.	What is Electro Encephalogram (EEG)? Discuss the types of Brain activity detected with EEG. Write any three applications.	10	L3	CO2
	b.	What are Pace Makers? Explain basic design and construction of Pace Makers.	10	L2	CO2
OR					
Q.6	a.	Justify Lungs as purification system.	10	L3	CO2
	b.	Explain architecture of Rod and Core cells with suitable diagram.	10	L2	CO2
Module – 4					
Q.7	a.	What is ultrasonography? Explain the uses and working principle.	10	L2	CO3
	b.	What is lotus leaf effect? Explain the mechanism and applications of super Hydrophobic effect.	10	L2	CO3
OR					
Q.8	a.	The structure and design of Kingfisher beak lead to the design of Bullet trains. Explain.	10	L2	CO3
	b.	Explain the working and applications of Bionic Leaf Technology.	10	L2	CO3

Module – 5					
Q.9	a.	Explain the use of Electrical tongue in food science.	10	L2	CO4
	b.	Explain the advantages and limitations of Artificial Intelligence for disease diagnosis.	10	L2	CO4
OR					
Q.10	a.	Explain Bioengineering solutions for muscular dystrophy and Osteroporosis.	10	L2	CO4
	b.	Explain most commonly used Bioprinting Techniques.	10	L2	CO4

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BBOK407

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Biology for Engineers

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is Cell? Explain Prokaryotic cells and Eukaryotic cell in brief.	10	L2	CO1
	b.	With a neat figure, explain the structure of plant cell and animal cell.	10	L2	CO1
OR					
Q.2	a.	What is a biomolecule? Explain the classification of biomolecule.	10	L1	CO1
	b.	Explain the properties and functions of Enzymes.	10	L2	CO1
Module – 2					
Q.3	a.	What are Bioplastics? Compare the properties of PHA and PLA as bioplastics.	10	L1	CO1
	b.	With an example, explain the development of DNA vaccines.	10	L3	CO2
OR					
Q.4	a.	Explain the importance of lipids and its application in cleaning agents.	10	L2	CO1
	b.	Discuss the development of vaccine for Covid 19.	10	L3	CO2
Module – 3					
Q.5	a.	Discuss about the Human Brain as a CPU system.	10	L3	CO2
	b.	What is ECG? Describe the various parts of ECG.	10	L2	CO1
OR					
Q.6	a.	Describe the kidney as a filtration system.	10	L3	CO2
	b.	What are Pacemakers? Briefly explain the various kinds of pacemakers.	10	L2	CO2
Module – 4					
Q.7	a.	What is Echolocation? Discuss the application of echolocation in Ultrasonography.	10	L1	CO2
	b.	Explain the structure and design of Kingfisher's beak led to bullet trains.	10	L3	CO2
OR					
Q.8	a.	Discuss about the human blood substitutes.	10	L1	CO1
	b.	What is a lotus effect? Explain the mechanism and applications of lotus leaf effect.	10	L4	CO2

Module – 5					
Q.9	a.	Explain the architecture of Muscular and Skeletal systems as Scaffolds.	10	L3	CO4
	b.	Discuss the applications of Artificial intelligence in the diagnosis of disease.	10	L4	CO4
OR					
Q.10	a.	What is Bioprinting? Discuss the process and applications of Bio printing.	10	L4	CO4
	b.	Write a note on : (i) Electrical Tongue (ii) Self-healing Bioconcrete.	10	L3	CO4

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Third Semester B.E./B.Tech. Degree Examination, June/July 2024

Mathematics for Computer Science

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. VTU Formula Hand Book is permitted.

3. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C																
Q.1	a.	Obtain the mean and variance of Poisson distribution.	06	L2	CO2																
	b.	Out of 800 families with 4 children each, how many families would be expected to have (i) 2 boys and 2 girls (ii) atleast one boy (iii) at most 2 girls. Assume equal probabilities for boys and girls.	07	L3	CO2																
	c.	The length of telephone conversation in a booth has been an exponential distribution and found on an average to be 5 minutes. Find the probability that a random call made from this booth (i) ends less than 5 minutes (ii) between 5 and 10 minutes.	07	L2	CO2																
OR																					
Q.2	a.	The probability distribution of a finite random variable X is given by <table><tr><td>X</td><td>-2</td><td>-1</td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>P(X)</td><td>0.1</td><td>k</td><td>0.2</td><td>2k</td><td>0.3</td><td>k</td></tr></table> (i) Find the value of k (ii) Variance (iii) P(x ≤ 2)	X	-2	-1	0	1	2	3	P(X)	0.1	k	0.2	2k	0.3	k	06	L2	CO1		
X	-2	-1	0	1	2	3															
P(X)	0.1	k	0.2	2k	0.3	k															
	b.	The number of accidents in a year to taxi drivers in a city follows a Poisson distribution with mean 3. Out of 1000 taxi drivers find approximately number of drivers with (i) more than 3 accidents in a year (ii) at most 2 accidents in a year.	07	L3	CO2																
	c.	The marks of 1000 students in an exam follows normal distribution with mean 70 and standard deviation 5. Find the students whose marks will be (i) less than 65 (ii) between 65 and 75. A(1) = 0.3413.	07	L3	CO2																
Module – 2																					
Q.3	a.	Given the following joint distribution of the random variables X and Y. Find the corresponding marginal distribution. Also compute the covariance. <table><tr><td>X \ Y</td><td>1</td><td>3</td><td>9</td></tr><tr><td>2</td><td>1/8</td><td>1/24</td><td>1/12</td></tr><tr><td>4</td><td>1/4</td><td>1/4</td><td>0</td></tr><tr><td>6</td><td>1/8</td><td>1/24</td><td>1/12</td></tr></table>	X \ Y	1	3	9	2	1/8	1/24	1/12	4	1/4	1/4	0	6	1/8	1/24	1/12	06	L3	CO2
X \ Y	1	3	9																		
2	1/8	1/24	1/12																		
4	1/4	1/4	0																		
6	1/8	1/24	1/12																		
	b.	A salesmen’s territory consists of 3 cities A, B and C. He never sells in the same city for 2 consecutive days. If he sells in city A, then the next day he sells in city B. However if he sells in either B or C then the next day he is twice as likely to sell in city A as in the other city. In the long run how often does he sell in each of the cities.	07	L3	CO3																
	c.	Show that $P = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1/2 & 1/2 & 0 \end{bmatrix}$ is a regular stochastic matrix. Also find the associated unique fixed probability vector.	07	L2	CO2																

OR					
Q.4	a.	Define probability vector, regular stochastic matrix, fixed prob vector.	06	L1	CO3
	b.	The joint probability distribution of two discrete random variables X and Y is $f(x, y) = k(2x + y)$, where x and y are integers such that $0 \leq x \leq 2$, $0 \leq y \leq 3$. i) Find the value of the constant k. ii) Show that the random variables X and Y are dependent iii) Find $P(X \geq 1, Y \leq 2)$.	07	L3	CO2
	c.	A fair coin is tossed thrice. The random variables X and Y are defined as X = 0 or 1 according as head or tail occurs on the first toss, y-number of heads. Compute $e(X, Y)$	07	L3	CO2
Module – 3					
Q.5	a.	Define statistical hypothesis, null hypothesis, Type-I error and Type-II error.	06	L1	CO4
	b.	In 324 throws of a six faced die an even number turned up 181 times. Is it reasonable to think that the 'die' is an unbiased one at 99% level?	07	L2	CO4
	c.	Before an increase in excise duty on tea, 800 people out of sample of 1000 were consumers of tea. After the increase in duty, 800 people were consumers of tea in a sample of 1200 persons. Find whether there is significant decrease in the consumption of tea after the increase in duty at 1%. (One tailed test at 1% is 2.33).	07	L3	CO4
OR					
Q.6	a.	A coin is tossed 1000 times and head turns up 540 times. Decide on the hypothesis that the coin is unbiased.	06	L2	CO4
	b.	In an exit poll enquiry it was revealed that 600 voters in one locality and 400 voters from another locality favoured 55% and 48% respectively a particular party to come to power. Test the hypothesis that there is a difference in the locality in respect of the opinion.	07	L3	CO4
	c.	A random sample for 1000 workers in company has mean wage of Rs.50 per day and standard deviation of Rs.15. Another sample of 1500 workers from another company has mean wage of Rs.45 per day and standard deviation of Rs.20. Does the mean rate of wages varies between the two companies at 95% confidence limit?	07	L3	CO4
Module – 4					
Q.7	a.	The mean life time of a sample of 25 bulbs is found as 1550 hrs with standard deviation of 120 hours. The company manufacturing the bulbs claims that the average life of their bulbs is 1600 hrs. Is the claim acceptable at 5% level of significance?	06	L3	CO4
	b.	The two independent samples of eight and seven items respectively had the following values of the variable: Sample 1 9 11 13 11 15 9 12 14 Sample 2 10 12 10 14 9 8 10 Do the two estimates of population variance differ significantly at 5% level of significance? F at 5% ($V_1 = 7, V_2 = 6$) = 4.21.	07	L3	CO4
	c.	Table gives the number of aircraft accidents that occurred during the various days of a week. Test whether the accidents are uniformly distributed over the week. $\chi^2_{5\%} (\gamma = 5) = 11.07$. Day Mon Tue Wed Thur Fri Sat Number of accidents 15 19 13 12 16 15	07	L3	CO4

OR																																														
Q.8	a.	Two random samples gave the following data: <table><tr><td></td><td>Size</td><td>Mean</td><td>Variance</td></tr><tr><td>Sample 1</td><td>8</td><td>9.6</td><td>1.2</td></tr><tr><td>Sample 2</td><td>11</td><td>16.5</td><td>2.5</td></tr></table> Can we conclude that the two samples have been drawn from the same normal population? $F_{5\%}(10, 7) = 3.64$.											Size	Mean	Variance	Sample 1	8	9.6	1.2	Sample 2	11	16.5	2.5	06	L2	CO4																				
	Size	Mean	Variance																																											
Sample 1	8	9.6	1.2																																											
Sample 2	11	16.5	2.5																																											
	b.	The following data relate to the marks obtained by 11 students in two tests. Second test is after intense coaching. Do the data indicate that the students have benefited by coaching? <table><tr><td>Test 1</td><td>19</td><td>23</td><td>16</td><td>24</td><td>17</td><td>18</td><td>20</td><td>18</td><td>21</td><td>19</td><td>20</td></tr><tr><td>Test 2</td><td>17</td><td>24</td><td>20</td><td>24</td><td>20</td><td>22</td><td>20</td><td>20</td><td>18</td><td>22</td><td>19</td></tr></table> ($t_{5\%}(\gamma = 10)$ is 1.81)										Test 1	19	23	16	24	17	18	20	18	21	19	20	Test 2	17	24	20	24	20	22	20	20	18	22	19	07	L3	CO4								
Test 1	19	23	16	24	17	18	20	18	21	19	20																																			
Test 2	17	24	20	24	20	22	20	20	18	22	19																																			
	c.	The mean value of a random sample of 60 items was found to be 145 and standard deviation is 40. Find the 95% confidence limits for the population mean.										07	L2	CO5																																
Module – 5																																														
Q.9	a.	The following figures relate to production in kgs of three variables A, B, C of wheat sown on 12 plots. <table><tr><td>A</td><td>14</td><td>16</td><td>18</td></tr><tr><td>B</td><td>14</td><td>13</td><td>15</td><td>22</td></tr><tr><td>C</td><td>18</td><td>16</td><td>19</td><td>19</td><td>22</td></tr></table> Apply one-way Anova using a 0.05 significance level in the production of the varieties. F_c at 5% (2, 9) d.f is 4.26.										A	14	16	18	B	14	13	15	22	C	18	16	19	19	22	10	L3	CO6																	
A	14	16	18																																											
B	14	13	15	22																																										
C	18	16	19	19	22																																									
	b.	Analyze and interpret the following statistics concerning output of wheat per field obtained as a result of experiment conducted to test four varieties of wheat viz., A, B, C and D under a Latin - Square design. <table><tr><td>C</td><td>B</td><td>A</td><td>D</td></tr><tr><td>25</td><td>23</td><td>20</td><td>20</td></tr><tr><td>A</td><td>D</td><td>C</td><td>B</td></tr><tr><td>19</td><td>19</td><td>21</td><td>18</td></tr><tr><td>B</td><td>A</td><td>D</td><td>C</td></tr><tr><td>19</td><td>14</td><td>17</td><td>20</td></tr><tr><td>D</td><td>C</td><td>B</td><td>A</td></tr><tr><td>17</td><td>20</td><td>21</td><td>15</td></tr></table>										C	B	A	D	25	23	20	20	A	D	C	B	19	19	21	18	B	A	D	C	19	14	17	20	D	C	B	A	17	20	21	15	10	L3	CO6
C	B	A	D																																											
25	23	20	20																																											
A	D	C	B																																											
19	19	21	18																																											
B	A	D	C																																											
19	14	17	20																																											
D	C	B	A																																											
17	20	21	15																																											
OR																																														
Q.10	a.	Four doctors each test four treatments for a certain disease and observe the number of days each patient takes to recover. The results are as follows: <table><tr><td>Doctor/Treatment</td><td>T₁</td><td>T₂</td><td>T₃</td><td>T₄</td></tr><tr><td>D₁</td><td>10</td><td>14</td><td>19</td><td>20</td></tr><tr><td>D₂</td><td>11</td><td>15</td><td>17</td><td>21</td></tr><tr><td>D₃</td><td>9</td><td>12</td><td>16</td><td>19</td></tr><tr><td>D₄</td><td>8</td><td>13</td><td>17</td><td>20</td></tr></table> Discuss the difference between doctors and treatments $F_{at 5\% \text{ level}}(3, 9)$ is 3.86.										Doctor/Treatment	T ₁	T ₂	T ₃	T ₄	D ₁	10	14	19	20	D ₂	11	15	17	21	D ₃	9	12	16	19	D ₄	8	13	17	20	10	L3	CO6							
Doctor/Treatment	T ₁	T ₂	T ₃	T ₄																																										
D ₁	10	14	19	20																																										
D ₂	11	15	17	21																																										
D ₃	9	12	16	19																																										
D ₄	8	13	17	20																																										

b.	A study of the effect of different types of anesthesia on the length of post-operative hospital stay yielded for the following for cesarean patients. Group 'A' was given an epidural MS providing additional safety. Group 'B' was given an epidural and Group 'C' was given a spinal is considered to be less dangerous and Group 'D' was given general anesthesia is considered to be the most dangerous. Note that the data are in the form of distribution for each group. <table><tr><td></td><td>Length of Stay</td><td>Number of patients</td></tr><tr><td rowspan="2">Group A</td><td>3</td><td>6</td></tr><tr><td>4</td><td>14</td></tr><tr><td rowspan="2">Group B</td><td>4</td><td>18</td></tr><tr><td>5</td><td>2</td></tr><tr><td rowspan="3">Group C</td><td>4</td><td>10</td></tr><tr><td>5</td><td>9</td></tr><tr><td>6</td><td>1</td></tr><tr><td rowspan="2">Group D</td><td>4</td><td>8</td></tr><tr><td>5</td><td>12</td></tr></table> Test for the existence of an effect due to anesthesia type at 0.01. $F_{0.01} = 4.13$		Length of Stay	Number of patients	Group A	3	6	4	14	Group B	4	18	5	2	Group C	4	10	5	9	6	1	Group D	4	8	5	12	10	L3	CO6
	Length of Stay	Number of patients																											
Group A	3	6																											
	4	14																											
Group B	4	18																											
	5	2																											
Group C	4	10																											
	5	9																											
	6	1																											
Group D	4	8																											
	5	12																											

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Third Semester B.E./B.Tech. Degree Examination, June/July 2024

Digital Design and Computer Organization

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Reduce the following Boolean expressions to the minimum number of literals. i) $x(x' + y)$ ii) $x + x'y$ iii) $(x + y)(x + y')$ iv) $xy + x'z + yz$ v) $(x + y)(x' + z)(y + z)$	10	L3	CO1
	b.	Determine the minimum SOP form using Karnaugh Map $F = A'B'C' + B'CD' + A'BCD' + AB'C'$.	10	L3	CO1
OR					
Q.2	a.	Simplify the Boolean function $F(w, x, y, z) = \Sigma(1, 3, 7, 11, 15)$ which has the don't care conditions $d(w, x, y, z) = \Sigma(0, 2, 5)$	10	L3	CO1
	b.	Simplify and implement the following Boolean function using NAND gates $F(x, y, z) = (1, 2, 3, 4, 5, 7)$.	10	L3	CO1
Module – 2					
Q.3	a.	Implement the design of combinational circuit BCD and excess 3 code converter.	10	L2	CO2
	b.	Implement full adder circuit using 3:8 decoders.	10	L2	CO2
OR					
Q.4	a.	With Truth table and K-map simplification, implement the full adder with basic gates and using two half adders an OR gate.	10	L2	CO2
	b.	Realize the Boolean function using 8:1 multiplexer $F(A, B, C, D) = \Sigma(1, 3, 4, 11, 12, 13, 14, 15)$.	10	L2	CO2
Module – 3					
Q.5	a.	Explain Bus structure with diagram, explain how different peripherals connected to the bus.	10	L2	CO3
	b.	Explain in detail about the word alignment of a machine (microprocessor based systems) what is the consecutive addresses of aligned words for 16, 32 and 64 bit word length of the machine? Give consecutive address for each of the following specified above.	10	L2	CO3
OR					
Q.6	a.	Write a note on : i) Register Transfer Notation (RTN) ii) Assembly Language Notation.	10	L2	CO3

	b.	Illustrate an indexed addressing mode with a assembly language program to find the sum of the Test 1, Test 2 and Test 3 scores of the N number of students.	10	L2	CO3
Module – 4					
Q.7	a.	Explain Hardware interrupt, enabling/disabling of interrupts and sequence of events in handling interrupt request from a single device.	10	L2	CO4
	b.	Explain memory mapped I/O and I/O interface for an input device with a diagram.	10	L2	CO4
OR					
Q.8	a.	Describe DMA with its register and controllers.	10	L2	CO4
	b.	Explain the effect of size, cost and speed in memory Hierarchy.	10	L2	CO4
Module – 5					
Q.9	a.	Explain the process of Fetching word from memory in processor.	10	L2	CO5
	b.	With a diagram, explain the single bus organization of the data path inside a processor.	10	L2	CO5
OR					
Q.10	a.	Describe how an ALU perform on arithmetic and logic operation along with input gating diagrams.	10	L2	CO5
	b.	Explain the complete set of operations involved in executing the instruction Add (R ₃) R ¹ along with control sequence.	10	L2	CO5

Third Semester B.E./B.Tech. Degree Examination, June/July 2024

Operating Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C																				
Q.1	a.	Explain multi programming and time sharing systems.	07	L3	CO1																				
	b.	Explain the dual mode operation in operating systems with a neat block diagram.	07	L3	CO1																				
	c.	What are virtual machines? Explain with a neat figure.	06	L3	CO1																				
OR																									
Q.2	a.	What are system calls? Briefly explain different types of system calls.	07	L3	CO1																				
	b.	List and explain the services provided by OS for the user in efficient operation of a system.	07	L3	CO1																				
	c.	What are micro kernels? With a neat figure, explain the micro kernel structure? Point out their advantages over layered approach.	06	L3	CO1																				
Module – 2																									
Q.3	a.	What is process? Explain different states of the process with state transition diagram and process control block.	08	L2	CO2																				
	b.	What is Interprocess communication? Explain.	06	L2	CO2																				
	c.	What is thread? How it is different from process? Discuss various multithreading models with suitable illustration.	06	L2	CO2																				
OR																									
Q.4	a.	Consider the following processes where smaller the number has higher priority. Draw the Gantt chart compute the waiting time and average turnaround time by using FCFS, SRTF, preemptive priority scheduling. <table border="1"><thead><tr><th>Processes</th><th>Arrival times</th><th>Burst time</th><th>Priority</th></tr></thead><tbody><tr><td>P₁</td><td>0</td><td>7</td><td>4</td></tr><tr><td>P₂</td><td>3</td><td>5</td><td>2</td></tr><tr><td>P₃</td><td>3</td><td>3</td><td>6</td></tr><tr><td>P₄</td><td>5</td><td>5</td><td>3</td></tr></tbody></table>	Processes	Arrival times	Burst time	Priority	P ₁	0	7	4	P ₂	3	5	2	P ₃	3	3	6	P ₄	5	5	3	12	L2	CO2
	Processes	Arrival times	Burst time	Priority																					
P ₁	0	7	4																						
P ₂	3	5	2																						
P ₃	3	3	6																						
P ₄	5	5	3																						
	b.	Discuss the benefits of multithreaded programming. Explain the threading issues in detail.	08	L2	CO2																				
Module – 3																									
Q.5	a.	What are the requirements that must be satisfied by a solution to the critical section problem? Illustrate with an example the Peterson’s solution for critical section problem.	08	L3	CO3																				
	b.	What is critical section problem and solutions to the problem? How to solve using semaphores?	06	L3	CO3																				
	c.	Explain the classical bounded buffer problem of synchronization. Give the solution	06	L3	CO3																				
OR																									
Q.6	a.	What is dead lock? What are the necessary conditions for the deadlock to occur? How to recover from deadlocks.	10	L3	CO3																				

1 of 2

	b.	Assume that there are 5 processes P ₀ to P ₄ and 4 types of resources. At time T ₀ the system has following: <table><tr><th rowspan="2">Processes</th><th colspan="4">Allocation</th><th colspan="4">Max</th><th colspan="4">Available</th></tr><tr><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th><th>A</th><th>B</th><th>C</th><th>D</th></tr><tr><td>P₀</td><td>0</td><td>1</td><td>1</td><td>0</td><td>0</td><td>2</td><td>1</td><td>0</td><td>1</td><td>3</td><td>1</td><td>0</td></tr><tr><td>P₁</td><td>1</td><td>4</td><td>4</td><td>1</td><td>1</td><td>6</td><td>5</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P₂</td><td>1</td><td>3</td><td>6</td><td>5</td><td>2</td><td>3</td><td>6</td><td>6</td><td></td><td></td><td></td><td></td></tr><tr><td>P₃</td><td>0</td><td>6</td><td>3</td><td>2</td><td>0</td><td>6</td><td>5</td><td>2</td><td></td><td></td><td></td><td></td></tr><tr><td>P₄</td><td>0</td><td>0</td><td>1</td><td>4</td><td>0</td><td>6</td><td>5</td><td>6</td><td></td><td></td><td></td><td></td></tr></table> Apply the bankers algorithm to answer following: (i) What is the content of need matrix? (ii) Is the system in a safe state? (iii) If the request from P ₁ (2, 1, 1, 0) arrives can it be granted?	Processes	Allocation				Max				Available				A	B	C	D	A	B	C	D	A	B	C	D	P ₀	0	1	1	0	0	2	1	0	1	3	1	0	P ₁	1	4	4	1	1	6	5	2					P ₂	1	3	6	5	2	3	6	6					P ₃	0	6	3	2	0	6	5	2					P ₄	0	0	1	4	0	6	5	6					10	L3	CO3
Processes	Allocation				Max				Available																																																																																						
	A	B	C	D	A	B	C	D	A	B	C	D																																																																																			
P ₀	0	1	1	0	0	2	1	0	1	3	1	0																																																																																			
P ₁	1	4	4	1	1	6	5	2																																																																																							
P ₂	1	3	6	5	2	3	6	6																																																																																							
P ₃	0	6	3	2	0	6	5	2																																																																																							
P ₄	0	0	1	4	0	6	5	6																																																																																							
Module – 4																																																																																															
Q.7	a.	What is paging? Differentiate between paging and segmentation.	06	L3	CO4																																																																																										
	b.	What are TLB? Explain TLB in detail with a simple paging system and neat diagram.	08	L3	CO4																																																																																										
	c.	Given the memory partitions of 100K, 500K, 200K, 300K and 600K, apply first fit, best fit and worst fit algorithms to place 212K, 417K, 112K and 426K.	06	L3	CO4																																																																																										
OR																																																																																															
Q.8	a.	What is page fault? With a neat diagram, explain the steps in handling page fault.	08	L3	CO4																																																																																										
	b.	Illustrate how demand paging affects system performance. What is thrashing how it can be controlled?	06	L3	CO4																																																																																										
	c.	Consider the following sequence: 7, 0, 1, 2, 0, 3, 0, 4, 2, 3, 0, 3, 2, 1, 2, 0, 1, 7, 0, 1 Assuming frame size of 4, apply LRU, FIFO and optimal algorithm to find the page faults. Find out which algorithm is most efficiency.	06	L3	CO4																																																																																										
Module – 5																																																																																															
Q.9	a.	Explain various file attributes and operations of files.	06	L3	CO5 CO6																																																																																										
	b.	With a neat diagram, explain two level and tree structured directory structure.	08	L3	CO5 CO6																																																																																										
	c.	What is file? Explain the file mounting.	06	L3	CO5 CO6																																																																																										
OR																																																																																															
Q.10	a.	Give the following sequence: 95, 180, 34, 119, 11, 123, 62, 64 with the head initially at 50 and ending at track 199. What is the total disk travelled by the disk drum to satisfy request using FCFS, SSTF, LOOK and CLOOK algorithms.	12	L3	CO5 CO6																																																																																										
	b.	Explain the access matrix model of implementing protection on OS.	08	L3	CO5 CO6																																																																																										

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Third Semester B.E./B.Tech. Degree Examination, June/July 2024
Data Structures and Applications

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.

Module – 1			M	L	C																														
Q.1	a.	What is data structure? List and explain data structure operations.	08	L2	CO1																														
	b.	Discuss four dynamic memory allocation functions.	08	L2	CO1																														
	c.	With suitable example, discuss self-referential structures.	04	L2	CO1																														
OR																																			
Q.2	a.	What is sparse matrix? Give the triplet form for given matrix and also find its transpose. A = <table><tr><td></td><td>0</td><td>1</td><td>2</td><td>3</td></tr><tr><td>0</td><td>10</td><td>0</td><td>0</td><td>40</td></tr><tr><td>1</td><td>11</td><td>0</td><td>22</td><td>0</td></tr><tr><td>2</td><td>0</td><td>0</td><td>0</td><td>0</td></tr><tr><td>3</td><td>20</td><td>0</td><td>0</td><td>50</td></tr><tr><td>4</td><td>0</td><td>15</td><td>0</td><td>25</td></tr></table>		0	1	2	3	0	10	0	0	40	1	11	0	22	0	2	0	0	0	0	3	20	0	0	50	4	0	15	0	25	06	L3	CO2
		0	1	2	3																														
0	10	0	0	40																															
1	11	0	22	0																															
2	0	0	0	0																															
3	20	0	0	50																															
4	0	15	0	25																															
	b.	Explain ADT stack.	07	L2	CO2																														
	c.	Define Stack. Implement the operations of stack using arrays.	07	L1	CO2																														
Module – 2																																			
Q.3	a.	What is the advantage of circular queue over ordinary queue? Discuss the implementation of circular queue operations.	08	L2	CO2																														
	b.	Demonstrate multiple stacks and queues with suitable examples.	12	L2	CO2																														
OR																																			
Q.4	a.	Explain Linked Stacks and Queues operations.	10	L2	CO2																														
	b.	Give the C functions for the following on singly linked list with example: i) Insert a node at the beginning ii) Delete a node at the front iii) Display	10	L3	CO3																														
Module – 3																																			
Q.5	a.	Define linked list? Implement C function for the following circular Doubly linked list: i) Insert a node at the beginning ii) Delete a node at the end iii) Display	10	L3	CO3																														
	b.	Develop a function to delete a node whose information field is specified in singly linked list.	10	L3	CO3																														
OR																																			
Q.6	a.	What is a tree? With suitable example, define i) Complete binary tree ii) Degree of the tree iii) Level of a node	07	L2	CO4																														
	b.	List and explain representation of a binary tree?	07	L2	CO4																														
	c.	Explain In-threaded binary tree.	06	L2	CO4																														

Module – 4					
Q.7	a.	For the given data, draw a binary search tree. 100, 85, 45, 55, 110, 20, 70, 65	07	L3	CO4
	b.	List and explain the common operations of binary search tree.	07	L2	CO4
	c.	Explain about forests.	06	L2	CO2
OR					
Q.8	a.	Define graph. Explain graph abstract data types.	10	L2	CO4
	b.	Explain the elementary graph operations.	10	L2	CO4
Module – 5					
Q.9	a.	Define hashing. Explain types of hashing functions in detail.	10	L2	CO5
	b.	Explain static hashing and dynamic hashing in detail.	10	L2	CO5
OR					
Q.10		Write a short note on :			
	a.	Leftist trees	06	L2	CO4
	b.	Optimal binary search tree	07	L2	CO5
	c.	Priority queues	07	L2	CO2

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BCS306A

Third Semester B.E./B.Tech. Degree Examination, June/July 2024
Object Oriented Programming with Java

Time: 3 hrs.

Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.

2. M : Marks, L: Bloom's level, C: Course outcomes.

Module – 1			M	L	C
Q.1	a.	Differentiate two paradigms of programming.	5	L2	CO1
	b.	Explain the various bitwise and short circuit operators in Java.	8	L2	CO1
	c.	Write a Java program with a method to check whether a given number is prime or not.	7	L3	CO1
OR					
Q.2	a.	Explain various scopes of variables in Java.	5	L2	CO1
	b.	Explain the arithmetic compound assignment and Ternary operators in Java.	8	L2	CO1
	c.	Write a Java program to perform linear search on an array elements accepted from keyboard and key element also accepted from key board.	7	L3	CO1
Module – 2					
Q.3	a.	Explain method overloading in Java with examples.	8	L2	CO2
	b.	Design a stack class to hold maximum of N numbers with a constructor, push, POP and Display methods. Develop Java main method to illustrate stack operations.	12	L3	CO2
OR					
Q.4	a.	Explain the role of “this” keyword and “static” keyword in Java.	8	L2	CO2
	b.	Design a class called “Employee” with fields ID, Name and Salary. Write a suitable constructors a method to raise salary and a static method to display. The number of Employee objects. Write suitable Main method for illustration.	12	L3	CO2
Module – 3					
Q.5	a.	Explain the role of “Super” with example Java program.	6	L2	CO3
	b.	For any class and any method as an example, explain method overriding.	5	L2	CO3
	c.	Develop a Java program to create class called “Shape”. Create 3 sub classes : circle, triangle and square. Each class has 2 member function area () and draw (). Demonstrate polymorphism with a suitable main program.	9	L3	CO3
OR					
Q.6	a.	Explain the order of constructor execution in a multilevel class hierarchy.	6	L2	CO3
	b.	Define dynamic method dispatch and write a code snippet in Java to demonstrate.	5	L1	CO3

	c.	Develop Java program to create interface Resizable with methods resize width (int width) and resize height (int height) that allow object to be resized. Create a class Rectangle that implements This Interface.	9	L3	CO3
Module – 4					
Q.7	a.	Explain four categories of visibility for class members based on packages.	6	L2	CO4
	b.	Give the general form of an exception handling block and write a Java program to illustrate multiple catch classes.	7	L2	CO4
	c.	Write a custom exception in Java called “less marks” and raise This exception when marks entered by valuator in the range [30 – 34]	7	L3	CO4
OR					
Q.8	a.	With code snippets, explain mechanism to create and import a package in Java.	6	L2	CO4
	b.	Write a Java program to create chained exceptions with top-level exception is Null Pointer Exception and its cause Arithmetic Exception.	7	L3	CO4
	c.	Develop a Java program to create custom exception for Negative odd numbers.	7	L3	CO4
Module – 5					
Q.9	a.	Explain various methods of thread class in Java.	6	L2	CO5
	b.	Write a Java program to create 4 threads and each thread when run, will sleep for 500 milliseconds and print its name before “Before Quitting”.	8	L3	CO5
	c.	Explain the use of Type wrappers in Java with example.	6	L2	CO5
OR					
Q.10	a.	Explain is Alive () and join () methods of Thread with example code snippet.	6	L2	CO5
	b.	Write a Java program to create 4 Thread and each Thread generates random number and prints it and sleeps for 800 msec and quits.	8	L3	CO5
	c.	Explain the concept of autoboxing /unboxing in expressions and methods.	6	L2	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Analysis and Design of Algorithms

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	What is an algorithm? Explain the fundamentals of algorithmic problem solving.	10	L2	CO1
	b.	Develop an algorithm to search an element in an array using sequential search. Calculate the best case, worst case and average case efficiency of this algorithm.	10	L3	CO1
OR					
Q.2	a.	Explain asymptotic notations with example.	10	L2	CO1
	b.	Give the general plan for analyzing the efficiency of the recursive algorithm. Develop recursive algorithm for computing factorial of a positive number. Calculate the efficiency in terms of order of growth.	10	L3	CO1
Module – 2					
Q.3	a.	Explain Strassen's matrix multiplication approach with example and derive its time complexity.	10	L3	CO2
	b.	What is divide and conquer? Develop the quick sort algorithm and write its best case. Make use of this algorithm to sort the list of characters: E, X, A, M, P, L, E.	10	L2	CO2
OR					
Q.4	a.	Distinguish between decrease & conquer and divide & conquer algorithm design techniques with block diagram. Develop insertion sort algorithm to sort a list of integers and estimate the efficiency.	10	L3	CO2
	b.	Define topological sorting. List the two approaches of topological sorting and illustrate with examples.	10	L2	CO2
Module – 3					
Q.5	a.	Define AVL tree with an example. Give worst case efficiency of operations on AVL tree. Construct an AVL tree of the list of keys: 5, 6, 8, 3, 2, 4, 7 indicating each step of key insertion and rotation.	10	L3	CO3
	b.	Define Heap. Explain the bottom-up heap construction algorithm. Apply heap sort to sort the list of numbers 2, 9, 7, 6, 5, 8 in ascending order using array representation.	10	L3	CO3
OR					
Q.6	a.	Define 2-3 tree. Give the worst case efficiency of operations on 2-3 tree. Build 2-3 tree for the list of keys 9, 5, 8, 3, 2, 4, 7 by indicating each step of key insertion and node splits.	10	L3	CO3
	b.	Design Horspool algorithm for string matching. Apply this algorithm to find the pattern BARBER in the text: JIM_SAW_ME_IN_A_BARBERSHOP	10	L3	CO3
Module – 4					
Q.7	a.	Apply Dijkstra's algorithm to find the single source shortest path for given graph [Fig.Q7(a)] by considering 's' as source vertex. Illustrate each step.	10	L3	CO4

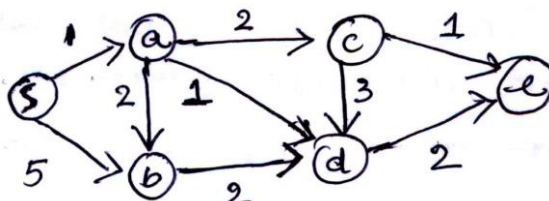


Fig.Q7(a)

- b.** Define transitive closure. Write Warshall's algorithm to compute transitive closure. Illustrate using the following directed graph.

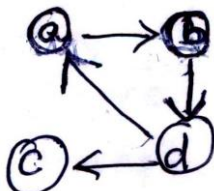


Fig.Q7(b)

OR

- Q.8 a.** Define minimum spanning tree. Write Kruskal's algorithm to find minimum spanning tree. Illustrate with the following undirected graph.

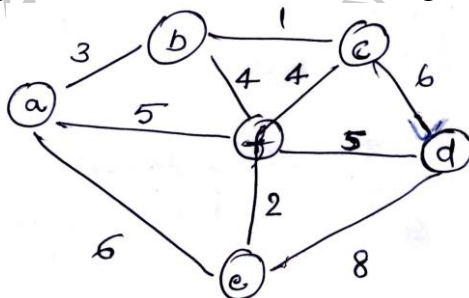


Fig.Q8(a)

- b.** Construct Huffman Tree and resulting code for the following:

Character	A	B	C	D	-
Probability	0.4	0.1	0.2	0.15	0.15

- (i) Encode the text : ABACABAD
(ii) Decode the text : 100010111001010

Module – 5

- Q.9 a.** Explain n-Queen's problem with example using backtracking approach.

- b.** Solve the following instance of the knapsack problem by the branch-and-bound algorithm. Construct state-space tree.

Item	Weight	Value
1	4	\$ 40
2	7	\$ 42
3	5	\$ 25
4	3	\$ 12

The knapsack's capacity W is 10.

OR

- Q.10 a.** Differentiate between Branch and Bound technique and Backtracking. Apply backtracking to solve the following instance of subset-sum problem $S = \{3, 5, 6, 7\}$ and $d = 15$. Construct a state space tree.

- b.** Explain greedy approximation algorithm to solve discrete knapsack problem.

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Microcontrollers

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.**2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Explain the architecture of an arm embedded device with a neat diagram.	10	L2	CO1
	b.	How are monitor and control internal operations performed in ARM core? Explain in brief.	10	L2	CO1
OR					
Q.2	a.	Explain memory management in ARM core. Compare cache and tightly coupled memory.	10	L2	CO1
	b.	Explain mechanism applied by ARM core to handle exception, interrupts using different vector table.	10	L2	CO1
Module – 2					
Q.3	a.	Examine data processing instructions requirement in the manipulation of data register? Explain in brief data processing instructions.	10	L2	CO2
	b.	Explain with examples the following 32-bit instruction of ARN processor i) CMN ii) MLA iii) MRS iv) BIC v) LDR.	10	L2	CO2
OR					
Q.4	a.	Explain the following with example : i) Stock operation ii) Swap instructions.	10	L2	CO2
	b.	Explain Branch instructions in ARM with suitable example. Demonstrate Branch instruct usage flow of execution with an example program.	10	L2	CO2
Module – 3					
Q.5	a.	How registers are allocated to optimize the program? Develop an assembly level program to find the sum of first to integer numbers.	10	L2	CO3
	b.	How compiler handles a “for loop” with variable number of iterations N and loop controlling with an example.	10	L2	CO3
OR					
Q.6	a.	Explain the following terms with an appropriate example : i) Pointer Aliasing ii) Portability issues.	10	L2	CO3
	b.	How function calling is efficiently used by ARM through APCS with an example program.	10	L2	CO3
Module – 4					
Q.7	a.	Explain ARM processors exception and modes with a neat diagram.	10	L2	CO4
	b.	Explain exception priorities and link register offset.	10	L2	CO4
OR					
Q.8	a.	List ARM firmware suite features. Explain firmware execution flow and Red Hat Boot.	10	L2	CO4
	b.	Explain IRQ and Fir exception, also to enable and disable IRQ and FIQ interrupts.	10	L2	CO4
Module – 5					
Q.9	a.	Explain basic architecture of cache memory.	10	L2	CO5
	b.	Explain process involved in main memory mapping to a cache memory.	10	L2	CO5
OR					
Q.10	a.	Explain with diagram set associative cache. How are efficiency is measured?	10	L2	CO5
	b.	Briefly explain cache line replacement policies with an example.	10	L2	CO5

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Database Management Systems

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C																												
Q.1	a.	Define database. Elaborate component modules of DBMS and their interactions.	10	L2	CO1																												
	b.	Describe the three-schema architecture. Why do we need mappings among schema levels?	06	L2	CO1																												
	c.	Explain the difference between logical and physical data independence.	04	L2	CO1																												
OR																																	
Q.2	a.	Draw an ER diagram for an COMPANY database with employee, department, project as strong entities and dependent as weak entity. Specify the constraints, relationships and ratios in the ER diagram.	10	L3	CO3																												
	b.	Define the following terms with example for each using ER notations: Entity, attribute, composite attribute, multivalued attribute, participation role.	10	L3	CO3																												
Module – 2																																	
Q.3	a.	Discuss the update operations and dealing with constraint violations with suitable examples.	08	L2	CO2																												
	b.	Illustrate the relational algebra operators with examples for select and project operation.	06	L2	CO2																												
	c.	Discuss the characteristics of relations that make them different from ordinary table and files.	06	L2	CO2																												
OR																																	
Q.4	a.	Perform (i) Student U instructor (ii) Student ∩ Instructor (iii) Student – Instructor (iv) Instructor – Student on the following tables: Student <table><tr><td>Fname</td><td>Lname</td></tr><tr><td>Susan</td><td>Yao</td></tr><tr><td>Ramesh</td><td>Shah</td></tr><tr><td>Johnny</td><td>Kohler</td></tr><tr><td>Barbara</td><td>Jones</td></tr><tr><td>Amy</td><td>Ford</td></tr><tr><td>Jimmy</td><td>Wang</td></tr><tr><td>Ernest</td><td>Gilbert</td></tr></table> Instructor <table><tr><td>Fname</td><td>Lname</td></tr><tr><td>John</td><td>Smith</td></tr><tr><td>Ricardo</td><td>Browne</td></tr><tr><td>Susan</td><td>Mao</td></tr><tr><td>Francis</td><td>Johnson</td></tr><tr><td>Ramesh</td><td>Shah</td></tr></table>	Fname	Lname	Susan	Yao	Ramesh	Shah	Johnny	Kohler	Barbara	Jones	Amy	Ford	Jimmy	Wang	Ernest	Gilbert	Fname	Lname	John	Smith	Ricardo	Browne	Susan	Mao	Francis	Johnson	Ramesh	Shah	04	L3	CO2
Fname	Lname																																
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John	Smith																																
Ricardo	Browne																																
Susan	Mao																																
Francis	Johnson																																
Ramesh	Shah																																
	b.	Consider the following relational database schema and write the queries in relational algebra expressions: EMP(<u>Eno</u> , Ename, Salary, Address, Phone, DNo) DEPT(<u>DNo</u> , Dname, DLoc, MgrEno) DEPENDENT(<u>Eno</u> , <u>Dep Name</u> , Drelation, Dage) (i) List all the employees who reside in ‘Belagavi’. (ii) List all the employees who earn salary between 30000 and 40000 (iii) List all the employees who work for the ‘Sales’ department (iv) List all the employees who have at least one daughter (v) List the department names along with the names of the managers	10	L3	CO2																												

	c.	Consider the two tables T ₁ and T ₂ shown below: T₁ <table border="1"><tr><th>P</th><th>Q</th><th>R</th></tr><tr><td>10</td><td>a</td><td>5</td></tr><tr><td>15</td><td>b</td><td>8</td></tr><tr><td>25</td><td>a</td><td>6</td></tr></table> T₂ <table border="1"><tr><th>A</th><th>B</th><th>C</th></tr><tr><td>10</td><td>b</td><td>6</td></tr><tr><td>25</td><td>c</td><td>3</td></tr><tr><td>10</td><td>b</td><td>5</td></tr></table> Show the results of the following operations: (i) $T_1 \bowtie_{T_1.P=T_2.A} T_2$ (ii) $T_1 \bowtie_{T_1.Q=T_2.B} T_2$ (iii) $T_1 \bowtie_{(T_1.P=T_2.A \text{ AND } T_1.R=T_2.C)} T_2$	P	Q	R	10	a	5	15	b	8	25	a	6	A	B	C	10	b	6	25	c	3	10	b	5	06	L3	CO2
P	Q	R																											
10	a	5																											
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25	a	6																											
A	B	C																											
10	b	6																											
25	c	3																											
10	b	5																											
Module – 3																													
Q.5	a.	Discuss the informal design guidelines for relation schema design.	08	L2	CO4																								
	b.	Define 1NF, 2NF, and 3NF with examples.	06	L2	CO4																								
	c.	Write the syntax for INSERT, UPDATE and DELETE statements in SQL and explain with suitable examples.	06	L2	CO3																								
OR																													
Q.6	a.	Discuss insertion, deletion and modification anomalies. Why are they considered bad? Illustrate with examples.	10	L2	CO3																								
	b.	Illustrate the following with suitable examples: (i) Datatypes in SQL (ii) Substring Pattern Matching in SQL.	10	L2	CO3																								
Module – 4																													
Q.7	a.	Consider the following relations: Student(<u>Snum</u> , Sname, Branch, level, age) Class(<u>Cname</u> , meet_at, room, fid) Enrolled(<u>Snum</u> , <u>Cname</u>) Faculty(<u>fid</u> , fname, deptid) Write the following queries in SQL. No duplicates should be printed in any of the answers. (i) Find the names of all Juniors (level = JR) who are enrolled in a class taught by I. Teach. (ii) Find the names of all classes that either meet in room R128 or have five or more students enrolled. (iii) For all levels except JR, print the level and the average age of students for that level. (iv) For each faculty member that has taught classes only in room R128, print the faculty member's name and the total number of classes she or he has taught. (v) Find the names of students not enrolled in any class.	10	L3	CO3																								
	b.	What do understand by correlated Nested Queries in SQL? Explain with suitable example.	04	L2	CO3																								
	c.	Discuss the ACID properties of a database transaction.	06	L2	CO4																								
OR																													
Q.8	a.	What are the views in SQL? Explain with examples.	04	L3	CO5																								
	b.	In SQL, write the usage of GROUP BY and HAVING clauses with suitable examples.	06	L2	CO3																								
	c.	Discuss the types of problems that may encounter with transactions that run concurrently.	10	L2	CO5																								

Module – 5					
Q.9	a.	What is the two phase locking protocol? How does it Guarantee serializability.	06	L2	CO5
	b.	Describe the wait-die and wound-wait protocols for deadlock prevention.	08	L2	CO5
	c.	List and explain the four major categories of NOSQL system.	06	L2	CO3
OR					
Q.10	a.	What is Multiple Granularity locking? How is it implemented using intension locks? Explain.	10	L2	CO5
	b.	Discuss the following MongoDB CRUD operations with their formats: (i) Insert (ii) Delete (iii) Read	06	L2	CO4
	c.	Briefly discuss about Neo4j data model.	04	L2	CO4

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Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Discrete Mathematical Structures

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks, L: Bloom's level, C: Course outcomes.*

Module – 1			M	L	C
Q.1	a.	Define tautology. Prove that for any propositions p, q, r the compound proposition. $[(p \wedge \neg q) \rightarrow r] \rightarrow [p \rightarrow (q \vee r)]$ is a tautology	06	L2	CO1
	b.	Test whether the following is a valid argument: If Ram studies then he will pass 12 th . If Ram passes 12 th then his father gifts him a bike. If Ram doesn't play video game then he will pass 12 th . Ram did not get a bike. $\therefore$ Ram played video game.	07	L3	CO1
	c.	Give direct proofs of the statements: i) If k and l are odd then k + l is even. ii) If k and l are odd then kl is odd.	07	L2	CO1
OR					
Q.2	a.	Define (i) Proposition (ii) Open statement (iii) Quantifiers	06	L2	CO1
	b.	Using the laws of logic, prove the following logical equivalence: $[(\neg p \vee \neg q) \wedge (F_0 \vee p) \wedge p] \Leftrightarrow p \wedge \neg q$	07	L2	CO1
	c.	Write the following statement in symbolic form and find its negation: “If all triangles are right angled then no triangle is equilateral”.	07	L2	CO1
Module – 2					
Q.3	a.	Prove by using mathematical induction. $1^2 + 2^2 + \dots + n^2 = \frac{n(n+1)(2n+1)}{6}$	06	L2	CO1
	b.	How many words can be made with or without meaning from the letters of the word “STATISTICS”? In how many of these a and c are adjacent? In how many vowels are together?	07	L3	CO2
	c.	Find the coefficient of x^3y^8 in the expansion of $(2x - y)^{11}$.	07	L2	CO2
OR					
Q.4	a.	Obtain the recursive definition for the sequence in each of the following cases: (i) $a_n = 5n$ (ii) $a_n = 3n + 7$ (iii) $a_n = n^2$ (iv) $a_n = 2 - (-1)^n$	06	L2	CO2
	b.	A woman has 11 close relations and wishes to invite 5 of them to dinner. In how many ways can she invite them if (i) there is no restriction on her choice. (ii) 2 persons will not attend separately (iii) 2 persons will not attend together.	07	L3	CO2
	c.	In how many ways can we distribute 7 apples and 5 oranges among 3 children such that each child gets atleast one apple and one orange?	07	L3	CO2

Module – 3					
Q.5	a.	State pigeon hole principle. Using pigeon hole principle find the minimum number of persons chosen so that atleast 5 of them will have their birthday in the same month.	06	L3	CO3
	b.	Let $A = \{a, b, c, d\}$ and $B = \{1, 2, 3, 4, 5\}$. Find the number of 1-1 functions and onto functions from (i) A to B (ii) B to A	07	L2	CO3
	c.	Let $A = \{1, 2, 3, 4, 5\}$. Define a relation R on $A \times A$ by $(x_1, y_1) R (x_2, y_2)$ iff $x_1 + y_1 = x_2 + y_2$. (i) Verify that R is an equivalence relation (ii) Determine the equivalence class of $[(2, 4)]$	07	L2	CO3
OR					
Q.6	a.	Consider the functions f and g from R to R defined by $f(x) = 2x + 5$ and $g(x) = \frac{1}{2}(x - 5)$. Prove that g is inverse of f.	06	L2	CO3
	b.	Let $A = \{1, 2, 3, 4\}$ and R be the relation on A defined by xRy if and only if $x < y$. Write down R as a set of ordered pairs. Write the relation matrix and draw the digraph. List out the in degrees and out degrees of every vertex.	07	L2	CO3
	c.	Let $A = \{1, 2, 3, 6, 9, 12, 18\}$ and define R on A by xRy iff 'x divides y'. Prove that (A, R) is a POSET. Draw the Hasse diagram for (A, R).	07	L2	CO3
Module – 4					
Q.7	a.	How many integers between 1 and 300 (inclusive) are divisible by (i) atleast one of 5, 6 or 8. (ii) None of 5, 6 and 8.	06	L3	CO4
	b.	At a restaurant 10 men handover their umbrellas to the receptionist, In how many ways can their umbrellas be returned so that (i) no man receives his own umbrella. (ii) atleast one gets his own umbrella. (iii) atleast two gets their own umbrellas.	07	L3	CO4
	c.	The number of virus affected files in a system is 1000 (to start with) and this increases by 250% every 2 hours. Use a recurrence relation to determine the number of virus affected files in the system after 12 hours.	07	L3	CO4
OR					
Q.8	a.	In how many ways one can arrange the letters of the word "CORRESPONDENTS" so that there are (i) no pair (ii) atleast 2 pairs of consecutive identical letters.	06	L3	CO4
	b.	4 persons P_1, P_2, P_3, P_4 who arrive late for a dinner party find that only one chair at each of five tables T_1, T_2, T_3, T_4 and T_5 is vacant. P_1 will not sit at T_1 or T_2 . P_2 will not sit at T_2 . P_3 will not sit at T_3 or T_4 . P_4 will not sit at T_4 or T_5 . Find the number of ways they can occupy the vacant chairs.	07	L3	CO4
	c.	Solve the recurrence relation $a_n - 6a_{n-1} + 9a_{n-2} = 0$ for $n \geq 2$ with $a_0 = 5, a_1 = 12$.	07	L2	CO4
Module – 5					
Q.9	a.	If * is an operation on Z defined by $xy = x + y + 1$, prove that $(Z, *)$ is an abelian group.	06	L2	CO5
	b.	Explain Klein-4 group with example.	07	L2	CO5
	c.	State and prove Lagrange's theorem.	07	L2	CO5
OR					
Q.10	a.	Prove that intersection of two subgroups of a group G is also a subgroup of G.	06	L2	CO5
	b.	Prove that $(Z_4, +)$ is a cyclic group. Find all its generators.	07	L2	CO5
	c.	Let $G = S_4$ for $\alpha = \begin{pmatrix} 1 & 2 & 3 & 4 \\ 2 & 3 & 4 & 1 \end{pmatrix}$ Find the subgroup $H = \langle \alpha \rangle$ determine the left cosets of H in G.	07	L3	CO5

Fourth Semester B.E./B.Tech. Degree Examination, June/July 2024

Graph Theory

Time: 3 hrs.

Max. Marks: 100

*Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.*

Module – 1				M	L	C
Q.1	a.	Define graph. List and explain the types of graph.		08	L1	CO1
	b.	Prove that the number of vertices of odd degree in a graph is always even.		06	L2	CO1
	c.	Define isomorphic graph and verify the following graphs are isomorphic or not. [Refer Fig.Q1(c)]		06	L2	CO1
Fig.Q1(c)						
OR						
Q.2	a.	Explain the following graphs: (i) Bi-partile graph (ii) Sub graphs (iii) WALK (iv) Path		10	L1	CO1
	b.	Prove that a simple graph with n vertices and K components can have at most $(n - K)(n - K + 1)/2$ edges.		10	L2	CO1
Module – 2						
Q.3	a.	State and prove necessary condition of a graph to be a Euler graph.		10	L2	CO2
	b.	List and explain the different operations on graph.		10	L2	CO2
OR						
Q.4	a.	Define diagram. Find the indegree and outdegree of the following graph [Fig.Q4(a)].		08	L2	CO2
	b.	Illustrate the travelling salesman problem using a graph.		06	L2	CO2
	c.	List and explain different digraphs and binary relations.		06	L2	CO2
Fig.Q4(a)						
Module – 3						
Q.5	a.	Define a tree. Prove that in a graph G there is one and only one path between every pair of vertices, G is a tree.		06	L1	CO3

	b.	Explain the following: (i) Cut-edge (ii) Cut-vertex (iii) Cut-set	06	L1	CO3
	c.	Find and construct the following: (i) Minimum possible height of 11 vertex binary tree (ii) A binary tree for a given 11 such that the farthest vertex is as far as possible from the root that must have exactly 2 vertices at each level, except at zero level.	08	L2	CO3
OR					
Q.6	a.	Prove that every circuit has an even number of edges in common with any cut set.	10	L2	CO3
	b.	Prove that ring 50 m of any two cut-sets in a graph is either a third cut-set or an edge disjoint union of cut-sets.	10	L2	CO3
Module – 4					
Q.7	a.	Define the following: (i) Planar graph (ii) Embedding (iii) Non-planar (iv) Kuratowski's 2 graph	08	L2	CO4
	b.	Explain the simple observation mode relationship between planar graph and dual G^* .	08	L2	CO4
	c.	Write a note on path matrix.	04	L1	CO4
OR					
Q.8	a.	Prove that two graphs G_1 and G_2 are isomorphic if and only if their incidence matrices $A(G_1)$ and $A(G_2)$ differ only by permutations of rows and columns.	10	L2	CO5
	b.	Describe the observations that can be made about circuit matrix $B(G)$ 01 and graph G .	10	L2	CO5
Module – 5					
Q.9	a.	Prove that every tree with two or more vertices is 2 - chromatic.	10	L2	CO5
	b.	Explain the following for chromatic polynomial: (i) Finding a maximal independent set (ii) Finding all maximal independent set.	10	L2	CO5
OR					
Q.10	a.	Prove that the vertices of every planar graph can be properly colored with five colors.	10	L2	CO5
	b.	Explain the Greedy colouring algorithm.	10	L2	CO5
